24MAT121 - Ordinary Differential Equation (ODEs)

SYLLABUS:

Unit I - First order ODEs, modeling, direction fields, separable ODEs, Exact ODEs and Integrating Factors, Linear ODEs and modeling. (Section 1.1 - 1.5)

Unit II - Second order ODEs, Homogenious and non-homogenious linears DEs, modeling a spring-mass system, Euler-Cauchy eqns, Existence and uniqueness of sol^{n}. Wronskian, solution by undetermined coefficients, solution by variation of parameters (sections 2.1, 2.2, 2.4 - 2.10)

unit III - Homogenious & non-homogenious higher order linear ODEs, System of ODEs, Series solutions of ODEs, legendre's eqn and legendre polynomials. (Section 3.1, 3.2, 3.3, 4.1, 5.1, 5.2)

unit IV - Extended power series method, Frobenius method, Bessel function & its general solution. (Section 5.3 - 5.5)

Unit V - Laplace Transforms, First & Second shifting theorems, Dirac-delta functions of system of ODEs. (Sections 6.1, 6.4, 6.7)

Text Book

* 1) Erwin Kreyzig - Advanced Engineering & mathematics 10th Edtn.

2) W.E Boyce & Diprima - Elementary diff eqns. 7th edn.

References.

1) M.D. Raisanghania Ordinary & partial diff eqn.

2) Dennis Zill - A first Course in DE's

Evaluatn pattern.

Internal : External

50 : 40

50 → 30 (CA), 20 (Mid (50))

30 (CA) → 15 + 15

15 → 5, 5 (2 Assignments of 5 mark each)

15 → 5, 5 (2 class tests of each mark)

40 → End term (100)

Unit 1

First Order ODEs.

Basic concepts

* An ordinary differential eqn of first order is an eqn that contains y' (first derivative) $\left(\frac{dy}{dx}\right)$, y, x or constants.

eg:- $y' + \sin x = 2$ linear - power of all diff coef is 1

non-linear - have power like 2, 3 ...

First order - max differential is 1 (y')

Second order - max " is 2 (y'')

- $y' + \sin x = 2$
- $xy' + 2\sin x = \cos x$
- $y' + 2y = 0$

} linear first order ODE

* The general form of a linear 1st order ODE can be written as

- $f(x, y, y') = 0$ —①

eqn ① can be in the following form.

- $y' = f(x, y)$ —②

eqn ② is called the explicit form of a first order linear ODE. (in term of y')

eg:- $x^{-3}y' - 4y^2 = 0$

$y' = 4y^2x^3$. → explicit form.

Solution :-

In $\frac{dy}{dx}$ y is dependent variable.
x - independent variable,

* A function $y = h(x)$ is a soln to eqn (2) in an open interval I if it satisfies the eqn by replacing y by h & y' by h'

eg :- consider $xy' = -y$. verify $y = \frac{c}{x}$ a soln or not.

$$xy' = -y \xrightarrow{c/x}$$

$$y' = c \times -\frac{1}{x^2}$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(x^{-1}) = -x^{-2} = \frac{-1}{x^2}$$

$$x \times \frac{-c}{x^2} = \frac{-c}{x} = -y$$

$xy' = -\frac{c}{x}$ $y' = \frac{-y}{x}$

$y' = \frac{-c}{x^2}$ $\frac{dy}{dx} = -\frac{y}{x}$

$\frac{dy}{dx} = \frac{-c}{x^2}$ $\frac{dy}{y} = -\frac{1}{x}dx$

$y = \frac{-1}{x^2}$ $\int \frac{dy}{y} = \int -\frac{1}{x}dx$

Solution curve :-

Let $y' = \cos x$

$$\frac{dy}{dx} = \cos x$$

$$dy = \cos x \, dx$$

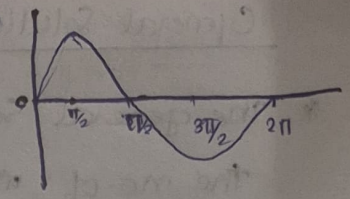
Integrating on both sides

$$\int dy = \int \cos x \, dx$$

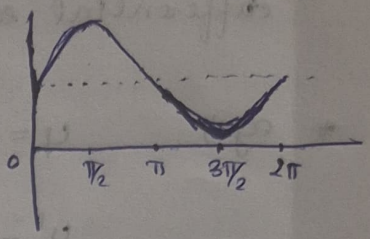
$$y = \sin x + C$$

* where c is an arbitrary constant. as C varies, we will get a family of solutions and each soln gives solution curves

* Soln curve when $C = 0$;

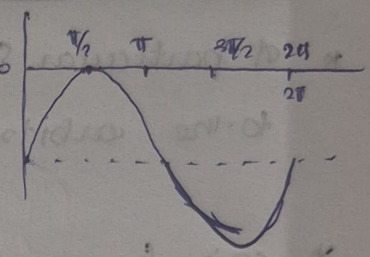


* Soln curve when $C = 1$;



by changing C value change occurs at value of y ; ∴ Vertical shift.

* Soln curve when $C = -1$;



Exponential growth & decay.

consider $y = ce^{3t}$ (c arg constant)

$y' = c \times 3e^{3t}$

$y' = 3ce^{3t}$ Since $y = ce^{3t}$,

$y' = 3y$ — ①

This implies y is a solution of ①.

This model of ODE is called the exponential **growth**.

→ $y' = ky$

* Similarly, $y' = -0.2y$ has a soln $y = ce^{-0.2t}$

hence this ODE model is called the exponential **decay**.

→ $y' = -ky$

$y' = -0.2y$

$\frac{dy}{dt} = -0.2y$

$\frac{dy}{y} = -0.2\,dt$

$\ln y = -0.2t + k$

$e^{\ln y} = -0.2t + k$

$y = e^{-0.2t}$

$e^k = c$

$y = e^{-0.2t} \cdot e^k$

$y = ce^{-0.2t}$

General Solution.

* The general soln of a differential eqn is that in which, the no. of arbitrary constants is equal to the order of the differential eqn.

* eg:- $y = Ce^{-0.2t}$ is a general soln to the 1st order ODE

$$y' = -0.2y$$

particular Solution.

* A particular soln is obtained by assigning some specific values to the arbitrary constants in the general soln.

* eg:- here; $y = 2e^{-0.2t}$ is a particular soln.

Initial Value Problem.

* An ODE together with an initial condition is called an IVP.

* it's general form;

$$y' = f(x, y) \quad \text{w.r.t} \quad y(x_0) = y_0$$

* when condition applied the ans we get always will be particular soln.

eg:- $y' = 3y$, $y(0) = 5.7$

$$\frac{dy}{dx} = 3y$$

$$\frac{dy}{y} = 3dx$$

$$\int \frac{1}{y} dy = 3x$$

$$\ln y = 3x + k$$

$$y = Ce^{3x}$$

applying given condn;

$$y(0) = 5.7$$

when $x=0$, $y = 5.7$

$$y = 5.7$$

$$C \cdot e^{3x} = 5.7$$

$$C \cdot e^{3x0} = 5.7$$

$$C \cdot 1 = 5.7$$

$$C = 5.7$$

∴ the particular solution

$$y = 5.7e^{3x}$$

- Modeling Problems.

1.Q:- Given an amount of a radioactive substance. say, 0.5g, find the amount present at any later time

Soln:- Let the amount present at any time is $y(t)$.

time rate of change $\frac{dy}{dt}$.

∴ $\frac{dy}{dt}$ ∝ amount present ($y(t)$) at any time (t).

$$\frac{dy}{dt} \propto y(t)$$

$$\frac{dy}{dt} = k\, y(t)$$

(k → proportionality constant)

$$\frac{dy}{dt} = ky$$

$$y' = ky$$

$$\int \frac{dy}{y} = \int k \cdot dt$$

$$\ln y = kt + k'$$

$$y = ce^{kt}$$

$$y = ce^{kt}$$

given at $t = 0$, $y = 0.5$

$$\therefore 0.5 = c \times e^{k \times 0} \qquad e^0 = 1$$

$$c = 0.5$$

$\therefore$ amount present at any time, $y(t) = 0.5\, e^{kt}$

2. Q:- The half life of a radioactive substance is the time in which half of the given amount disappears. What is the half life of Radium ($_{88}Ra^{226}$) if we take the proportionality constant $k = 1.4 \times 10^{-11}$

Soln: Let y_0 be the amount present at $t=0$. The ODE model is $y' = ky$

$$y = ce^{kt}$$

at $t = 0$, $y = y_0$

$$y_0 = ce^{k \cdot 0}$$

$$c = y_0$$

$$\therefore y = y_0 e^{kt}$$

at half life, $y_0 \rightarrow y_0/2$

to find t ?

at half life ;

$$\frac{y_0}{2} = y_0 e^{kt}$$

$$\frac{1}{2} = e^{kt}$$

$$\ln\left(\tfrac{1}{2}\right) = kt$$

$$\ln(1) - \ln(2) = -1.4 \times 10^{-11} \times t \quad \text{(decay)}$$

$$-\ln 2 = -1.4 \times 10^{-11} t$$

$$t = \frac{-\ln 2}{-1.4 \times 10^{-11}}$$

$$= 0.4951 \times 10^{11}$$

In years;

$$t = \frac{0.4951 \times 10^{11}}{365 \times 60 \times 60 \times 24} = 1570 \text{ year}$$

3.Q: The half life of Radium is about 3.6 days. Given 1g, how much will still be present after (a) 1 day? (b) 1 year?

Soln: Let at $t=0$, $y_0 = 1g$

$y' = ky$

$y = ce^{kt}$

$1 = c \cdot e^{k \times 0}$

$c = 1$

$y = e^{kt}$

$y = e^{k \times 3.6}$

3.6

$y = y_0 e^{kt}$

given $t = 3.6$

at half life;

$\frac{y_0}{2} = y_0 e^{kt}$

$\frac{1}{2} = e^{kt}$

$\ln(1/2) = k \times 3.6$

$3.6k = \ln(1/2)$

$k = \frac{\ln(0.5)}{3.6} = -0.1925$

$y(1) = 1 \times e^{-0.1925t}$

a) The amount after 1 day.

$y(1) = e^{-0.1925} = 0.8248\ g$

b) The amount after 1 year = 365 days

$y(365) = e^{-0.1925 \times 365} = 3.05 \times 10^{-31}\ g$

4:Q Find the curve through the point (1,1) in x-y plane such that each of its points, the slope is $-\frac{y}{x}$.

model: $\frac{dy}{dx} = \frac{-y}{x}$ Slope $= \frac{dy}{dx}$

$\int \frac{dy}{+y} = \int \frac{-1}{x} dx$

$+\ln y = -\ln x + K$

$y = k\frac{1}{x}$ - general curve.

At (1,1) $x=1, y=1$

$y = k\frac{1}{x}$

$1 = \frac{k}{1}$

$k = 1$

The curve; $y = 1/x$

5.Q :- A tank contains 800 gallons of water. 200 lbs of salt is dissolved. 2 gallons of fresh water runs in per minute and 2 gallons of the mixture in the tank kept uniform by stirring runs out per minute. How much salt is left in the tank after 5hrs.

$y_0 \Rightarrow 200$ lbs

$S(t)$ = amnt of salt (lb) in tank at t minutes.

Rate of enflow = 2 gal/min (fresh water)
Rate of outflow = 2 gal/min (mixture)

Vol of water remains constant = 800 gallons.

Rate of salt entering = 0
Rate of salt leaving; $S(t)$ / 800

Con. of salt = $\frac{S(t)}{800}$ outflow = 2 gal/min

Rate of salt leaving the tank

$= 2 \times \frac{x(t)}{800} = \frac{x(t)}{400}$

$\frac{ds}{dt}$ = Rate in − Rate out

$\frac{ds}{dt} = 0 - \frac{s}{400}$ $\therefore \frac{ds}{dt} = -\frac{s}{400}$

$\frac{1}{s} ds = -\frac{1}{400} dt$

$\ln s = -\frac{t}{400} + c$

$s = ce^{-t/400}$

$x(0) = 200$

$200 = ce^0 = c = 200$

$x(t) = 200e^{-t/400}$

$x(300) = 200e^{-300/400} = 94.5$ lb

Exact ODE

* a function $u(x,y)$ has continous partial derivatives, then the total differential is given by,

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \quad -(1)$$

if $u(x,y) = c$, $du = 0$

if $u(x,y) = x^2 + x^2 y^3 = c$, $du = (1 + 2x y^3) dx + 3y^2 x^2 dy = 0$

* A first order linear ODE can be written as

$$M dx + N dy = 0 \quad -(2)$$

eqn (2) is called Exact differential equation, if the differential form $M \cdot dx + N \cdot dy$ is exact.

ie, $M \cdot dx + N \cdot dy$ is the differential of a function $u(x,y)$.

$$\Rightarrow du = M dx + N dy \quad -(3)$$

from (2) ; $du = 0$

Integrating on both sides . . . .

$$u(x,y) = c$$

from (3) and (1)

$$M dx + N dy = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$\therefore$ $\boxed{M = \frac{\partial u}{\partial x}}$ & $\boxed{N = \frac{\partial u}{\partial y}}$ ← partial derivatives.

differentiating partially, first w.r.t y

$$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial u}{\partial y}$$

$$\frac{\partial^2 u}{\partial y\,\partial x} = \frac{\partial M}{\partial y} \quad -(a)$$

Similarly, w.r.t x

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial N}{\partial x}$$

$$\frac{\partial^2 u}{\partial y\,\partial x} = \frac{\partial N}{\partial x} \quad -(b)$$

Similarly,

$$\frac{\partial^2 u}{\partial y\,\partial x} = \frac{\partial^2 u}{\partial y\,\partial x}$$

(a) & (b)

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}}$$ ← condition for exactness of eqn (2).

we have,

$$\frac{\partial u}{\partial x} = M$$

integrating w.r.t x (y is constant)

$$u = \int M\,dx + K(y) \quad -(I)$$

↘ constant of int

Similarly,

$$\frac{\partial u}{\partial y} = N$$

int... w.r.t y (x is constant)

$$u = \int N\,dy + l(x) \quad -(II)$$

↘ const of int

① Q:- Check the exactness and hence solve the following equation.

$$\cos(x+y)\,dx + \left(3y^2 + 2y + \cos(x+y)\right)dy = 0$$

Soln :- In the form of $M\,dx + N\,dy = 0$

$$M = \cos(x+y) \qquad N = \left(3y^2 + 2y + \cos(x+y)\right)$$

x constant, y constant

$$\frac{\partial M}{\partial y} = -\sin(x+y) \qquad \frac{\partial N}{\partial x} = -\sin(x+y)$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \text{eqn is exact.}$$

↘ cond^n derived from exactness

Soln is;

$$u = \int M\,dx + K(y)$$

$$u = \int \cos(x+y)\,dx + K(y)$$

int w.r.t y, x const

$$u = \frac{\sin(x+y)}{1} + K(y)$$

$\int \cos ax\,dx = \frac{\sin ax}{a}$

$$u = \sin(x+y) + K(y)$$

Similarly,

differentiating partially w.r.t y

$$\frac{\partial u}{\partial y} = \cos(x+y) + k'(y)$$

from eqn ; N

$$N = \cos(x+y) + k'(y)$$

$$\therefore \quad 3y^2 + 2y + \cos(x+y) = \cos(x+y) + k'(y)$$

$$k'(y) = 3y^2 + 2y$$

Integrating w.r.t y

$$k(y) = \int 3y^2 + 2y \, dy + c$$

$$k(y) = y^3 + y^2 + c$$

$\therefore$ Solution from (a)

$$u(x,y) = \sin(x+y) + y^3 + y^2 + c$$

Since $u(x,y)$ is constant.

$$u(x,y) = k$$

$$k = \sin(x+y) + y^3 + y^2 + c$$

$$\sin(x+y) + y^3 + y^2 = c'$$

$$u = \int N\,dy + l(x)$$ int w.r.t y (x is const)

$$u = \int (3y^2 + 2y + \cos(x+y))\,dy + l(x)$$

$$u = y^3 + y^2 + \sin(x+y) + l(x)$$

diff w.r.t x, (y is constant)

$$\frac{\partial u}{\partial x} = 0 + 0 + \cos(x+y) + l'(x)$$

$$\frac{\partial u}{\partial x} = \cos(x+y) + l'(x)$$

$$M = \cos(x+y) + l'(x)$$

$$\cos(x+y) = \cos(x+y) + l'(x)$$

$$l'(x) = 0$$

Integrating w.r.t x ;

$$l(x) = c'$$

$$k = u(x,y) = y^3 + y^2 + \sin(x+y) + c'$$

$$y^3 + y^2 + \sin(x+y) = c$$

② Solve IVP $Mdx + Ndy = 0$

$(\cos y \sinh x + 1)dx - \sin y \cosh x\, dy = 0 \quad y(1) = 2$

Soln:-

$M = \cos y \sinh x + 1$

$N = -\sin y \cosh x$

$\frac{\partial M}{\partial y} = -\sin y \cdot \sinh x$

$\frac{\partial N}{\partial x} = -\sin y \cdot \sinh x$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$\therefore$ Eqn is exact.

The solution is .

$u = \int N dy + l(x)$

$u = \int -\sin y \cdot \cosh x\, dy + l(x)$

$u = \cos y \cdot \cosh x + l(x) \quad \text{—①}$

differentiating ① partially w.r.t x.

$\frac{\partial u}{\partial x} = \cos y \cdot \sinh x + l'(x)$

But $\frac{\partial u}{\partial x} = M$

$\cos y \cdot \sinh x + 1 = \cos y \cdot \sinh x + l'(x)$

$l'(x) = 1$

Integrating w.r.t x

$l(x) = x + c^*$

$\therefore$ the solution,

$k = u = \cos y \cosh x + x + c^*$

exact eqn from ? hyperbolic fn

$\cos y \cdot \cosh x + x = c$

$u(x, y) = c$

given $y(1) = 2$

when $x = 1, y = 2$

$\cos 2 \cdot \cosh 1 + 1 = c$

$\therefore$ the solution

$\cos y \cdot \cosh x + x = 1 + \cos 2 \cdot \cosh x 1$

$= \underline{0.358}$

Non-exactness.

Eg: Consider, $-y\,dx + x\,dy = 0$ ① $(M\,dx + N\,dy = 0)$

$M = -y$

$N = x$

$\frac{\partial M}{\partial y} = -1$

$\frac{\partial N}{\partial x} = 1$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

∴ Eqn is not exact.

Suitable $\int$

Let $F = 1/x^2$ (fn of x alone)

multiply F on both sides of ①

① $\Rightarrow$ $\frac{-y\,dx + x\,dy}{x^2} = 0$

Integrating ...

$d\left(\frac{y}{x}\right) = 0$

$\frac{y}{x} = c$

Here F is called the integrating factor (IF)

Theorem 1

Let $P\,dx + Q\,dy = 0$ be a non-exact eqn.

Let $R = \frac{1}{Q}\left[\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right]$ depends only on x

Then the integrating factor (IF), $F(x) = e^{\int R(x)\,dx}$

Theorem 2

Let $P\,dx + Q\,dy = 0$ be a non-exact eqn

Let $R^* = \frac{1}{P}\left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right]$ depends only on y,

the IF, $F^*(y) = e^{\int R^*(y)\,dy}$

Q: Check for exactness and hence, solve the following equations

$(e^{x+y} + ye^y)\,dx + (xe^y - 1)\,dy = 0 \; ; \; y(0) = -$

$P = e^{x+y} + ye^y$ $\quad \frac{\partial P}{\partial y} = e^{x+y} + ye^y + e^y$

$Q = xe^y - 1$ $\quad \frac{\partial Q}{\partial x} = e^y$

$\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$

∴ (not exact)

$$R = \frac{1}{Q}\left[\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right]$$

$$= \frac{1}{xe^y - 1}\left[e^{x+y} + ye^y + e^y - e^y\right)$$

$$R = \frac{1}{xe^y - 1}\left(e^{x+y} + ye^y\right)$$ not a fn of x alone.

$$R^* = \frac{1}{P}\left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right]$$

$$= \frac{1}{e^{x+y} + ye^y}\left(e^y - \left(e^{x+y} + ye^y + e^y\right)\right)$$

$$= \frac{1}{e^{x+y} + ye^y}\left(-e^{x+y} + -ye^y\right)$$

$$= \frac{-(e^{x+y} + ye^y)}{e^{x+y} + ye^y}$$

$$R^* = -1$$

By theorem 2 ;

$$IF = e^{\int R^*(y)\,dy}$$

$$= e^{\int -1\,dy} = e^{-y}$$

Exact eqn is

$$e^{-y}\left\{(e^{x+y} + ye^y)dx + (xe^y - 1)dy\right\} = 0$$

$$\therefore (e^x + y)\,dx + (x - e^{-y})\,dy = 0$$

$$M\,dx + N\,dy = 0$$

$M = e^x + y$ $\quad \frac{\partial M}{\partial y} = 1$

$N = x - e^{-y}$ $\quad \frac{\partial N}{\partial x} = 1$ $\quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

the soln is ;

$$u = \int M\,dx + k(y)$$

$$u = \int (e^x + y)\,dx + k(y)$$

$$u = e^x + yx + k(y) \quad \text{—①}$$

diff u partially w.r.t y.

$$\frac{\partial u}{\partial y} = e^x + x + k'(y)$$

$$\frac{\partial u}{\partial y} = N$$

$\therefore$

$x - e^{-y} = x_1 + k'(y)$

$k'(y) = -e^{-y}$

integrating...

$k(y) = e^{-y} + C^*$

Exact differential Equations.

eqn :- $M\,dx + N\,dy = 0$

cond^n for exactness : $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$\frac{\partial u}{\partial x} = M$

$\frac{\partial u}{\partial y} = N$

Solution :- 1) $u = \int M\,dx + k(y)$

2) $u = \int N\,dy + l(x)$

Non-exact differential eqn & IF

$Pdx + Qdy = 0$ is not exact

1) $IF = e^{\int R\,dx}$, $R = \frac{1}{Q}\left[\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right]$

2) $IF = e^{\int R^*\,dy}$, $R^* = \frac{1}{P}\left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right]$

PROBLEMS.

1] Find the general solution to the eqn

a) $2xy\,dx + x^2\,dy = 0$

ans:- $M = 2xy$ (given)

$N = x^2$

$\frac{\partial M}{\partial y} = \frac{d}{\partial y}(2xy)$ $\qquad \frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(x^2) = 2x$

$\frac{\partial M}{\partial y} = 2x$ $\qquad \frac{\partial N}{\partial y} = 2x$

$\therefore$ eqn is exact.

$\therefore$ to find the solution : ...

1) $u = \int M\,dx + k(y)$

$= \int 2xy \cdot dx + k(y)$

$= 2\frac{x^2}{2}\,y + k'(y)$

$u = x^2y + k(y) \rightarrow$ soln

But since $k(y)$ is unknown,

to find $k(y)$ ...

differentiating u w.r.t 'y'.

$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}(x^2y + k(y)) = x^2 \times 1 + k'(y)$

$= x^2 + k'(y)$

we have $\frac{\partial y}{\partial x} = M$ $\quad$ ✱

$$\frac{\partial u}{\partial y} = N$$

$$\therefore \quad \frac{\partial u}{\partial y} = N$$

$$2x^2 + k'(y) = x^2$$

$$k'(y) = 0$$

Integrating

$$\int k'(y) = \int 0$$

$$k(y) = c^*$$

↳ constant

$$u = yx^2 + c^*$$

$$\therefore \quad k = yx^2 + c^*$$

$$k - c^* = yx^2$$

another constant

$$yx^2 = k - c^* = c$$

↳ another constant

$$\therefore yx^2 = c$$

) $e^{x^2} 2xy\,dx + e^{x^2} dy = 0 \; ; \; y(0) = 2$ (IVP)

$$M = e^{x^2} 2xy$$

$$N = e^{x^2}$$

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(e^{x^2} \cdot 2xy) = e^{x^2} \times 2x = 2xe$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(e^{x^2}) = e^{x^2} \cdot 2x = 2xe^{x^2}$$

$$\therefore$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \therefore \text{ eqn is exact.}$$

1) ~~$u = \int M\,dy + k(y)$~~

2) $u = \int N\,dy + l(x)$

$$= \int e^{x^2} dy + l(x) \qquad \int dy = y$$

$$u = e^{x^2} \cdot y + l(x) \quad \text{---(1)} \cdot \text{ soln}$$

diff w.r.t. x

$$\frac{\partial u}{\partial x} = e^{x^2} \cdot 2x \cdot y + l'(x)$$

$$= 2xy \cdot e^{x^2} + l'(x)$$

$$\frac{\partial u}{\partial x} = M$$

$$2xy \cdot e^{x^2} + l'(x) = e^{x^2} 2xy$$

$$l'(x) = 0$$

Integrating

$$l'(x) = c^*$$

$u = e^{x^2} y + c^*$ — eqn ① becomes

$\therefore \quad k = u = e^{x^2} y + c^*$

$k - c^* = e^{x^2} y$

$\underline{\underline{ye^{x^2} = c}}$

Applying $y(0) = 2$ ↳ ie when $x = 0$, then $y = 2$

$2e^0 = c$

$c = 2$

$\therefore$ the partial sol^n

$\underline{\underline{ye^{x^2} = 2}}$

c) $(xy+1)\,dx + x^2 dy = 0 \quad , y(1) = 1$

$P = xy + 1$

$Q = x^2$

$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(xy+1) = x$

$\frac{\partial Q}{\partial x} = \frac{\partial (x^2)}{\partial x} = 2x$

$\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$ $\therefore$ eqn is not exact.

$R = \frac{1}{Q}\left[\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right]$

$= \frac{1}{x^2}\left[x - 2x\right]$

$= \frac{x - 2x}{x^2}$

$= \frac{1}{x} - \frac{2}{x}$

$= \underline{\underline{\frac{-1}{x}}}$ $\therefore$ fn of x alone

$IF = e^{\int R\,dx}$

$= e^{\int -1/x\,dx}$ $\qquad \int \frac{-1}{x}\,dx = -\ln x$

$= e^{-\ln x}$ $\qquad e^{-\ln x} = e^{\ln x^{-1}}$

$= e^{\ln x^{-1}}$

$= x^{-1}$

$= \underline{\underline{\frac{1}{x}}}$

$\therefore$ The exact eqn is

$\frac{1}{x}(xy+1)\,dx + \frac{1}{x} x^2 dy = 0$

$(y + 1/x)\,dx + x\,dy = 0$

The Sol^n.

$u = \int M\,dy + l(x) = \int x\,dy + l(x) = xy + l(x)$

diff w.r.t x

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}(xy + \ell n x)$$

$$= xy + c'(x)$$

$$\frac{\partial u}{\partial x} = M$$

$$y + \frac{1}{x} = y + c'(x)$$

$$\therefore c'(x) = \frac{1}{x}$$

Integrating Both sides

$$\therefore$$

$$c(x) = \ln x + c^*$$

$$K = u = xy + \ln x + c^*$$

$$K - c^* = xy + \ln x$$

$$xy + \ln x = C$$

given $y(1) = 1$

when $x = 1, y = 1$

$$1 \times 1 + \ln 1 = C$$

$$1 + \ln 1 = C$$

$$\ln 1 + 0 = C$$

$$C = 1$$

$$xy + \ln x = 1$$

d) $2x \tan y \, dx + \sec^2 y \, dy = 0 \; ; y(0) = 1$

$P = 2x \tan y \qquad \frac{\partial P}{\partial y} = 2x \cdot \sec^2 y$

$Q = \sec^2 y \qquad \frac{\partial Q}{\partial x} = 0$

$\therefore$ not exact. $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$

$$R = \frac{1}{Q}\left[\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right]$$

$$= \frac{1}{\sec^2 y}\left[2x \cdot \sec^2 y - 0\right)$$

$$= 2x$$

$\int R\,dx$
$= \int 2x\,dx$
$= x^2$

$$IF = e^{\int R\,dx}$$

$$= e^{\int 2x \cdot dy x}$$

$$= e^{x^2}$$

$\therefore$ The exact eqn ;

or $e^{x^2} \cdot 2x \tan y \, dx + e^{x^2} \sec^2 y \, dy = 0$

$\Downarrow M \qquad \Downarrow N$

Soln is ;

$$u = \int N\,dy + l(x)$$

$$= \int e^{x^2} \sec^2 y\,dy + l(x)$$

$u = e^{x^2}\tan y + l'(x)$

$k = e^{x^2}\tan y + l(x)$

$$\frac{\partial u}{\partial x} = e^{x^2} \cdot 2x \cdot \tan y + l'(x)$$

$$\frac{\partial u}{\partial x} = M$$

$$e^{x^2} \cdot 2x \cdot \tan y + l'(x) = e^{x^2} 2x \tan y$$

$$l'(x) = 0$$

Integ...

$$l(x) = c^*$$

$$k = e^{x^2} \tan y + c^*$$

$$k - c^* = e^{x^2} \cdot \tan y$$

$$e^{x^2} \cdot \tan y = c$$

given $y(0) = 1$

when $x = 0$, $y = 1$

$\therefore e^0 \cdot \tan(1) = c$

$c = \tan(1)$

$\therefore$ the soln ;

$$e^{x^2} \tan y = \tan(1)$$

Separable First Order ODE

Eqn ① Let $y' = f(x,y)$ —①

$$\frac{dy}{dx} = xy$$

$$\frac{dy}{y} = x\,dx$$

$$g(y)\,dy = f(x)\,dx$$

① can be written as

$$g(y)\,dy = f(x)\,dx \quad —②$$

Seperable form

The solution is obtained by integrating ② and is given by,

$$\int g(y)\,dy = \int f(x)\,dx + c$$

eg: 1. Solve $xy' = 1 + y^2$ by separating the variables.

$$\frac{dy}{dx} = 1 + y^2$$

$dy = y^2 + 1\,dx$

$$\frac{dy}{1+y^2} = dx$$

int... on both sides.

$$\int \frac{dy}{1+y^2} = \int dx + c$$

$\tan^{-1} y = x + c$

$y = \tan(x + c)$

eg: 2) $y' = -2xy$; $y(1) = 1.8$

$\frac{dy}{dx} = -2xy$

$\frac{dy}{y} = -2x\,dx$

integrating both sides.

$\int \frac{dy}{y} = \int -2x\,dx + c$

$\ln y = -x^2 + c$

Apply the given cond^n.

when $x = 1$, $y = 1.8$

$\ln(1.8) = -(1)^2 + c$

$c = -\ln(1.8)$

$\ln(1.8) = -1 + c$

$c = 1 + \ln(1.8)$

$c = 1.959$

∴ The sol^n is, $\ln y = -x^2 + 1.959$

$\int \frac{1}{\ln y} = \tan^{-1} y$

∴ $y = e^{-x^2 + 1.959}$

$y = e^{-x^2} \cdot e^{1.959}$

$y = 4.9e^{-x^2}$

Extended method : Reduction to seperable form.

Let $y' = f(x, y)$ —①

If eqn ① is not in seperable form, then give a substitution.

$y = ux$ $\left(u = \frac{y}{x}\right)$

(pdt rule)

Then, $y' = u'x + u$

By substituting and y' in ①, the eqn ① become in separable form.

eg④ Solve $2xyy' = y^2 - x^2$ —①

put $y = ux$, $y' = u'x + u$

① becomes,

$y' = \frac{y^2 - x^2}{2x \cdot y}$

$y' = \frac{y}{2x} - \frac{x}{2y}$

$y' = \frac{1}{2}\left(\frac{y}{x}\right) - \frac{1}{2}\left(\frac{1}{y/x}\right)$

$= \frac{1}{2}\left(\frac{ux}{x}\right) - \frac{1}{2}\left(\frac{1}{ux/x}\right)$

$y' = \frac{u}{2} - \frac{1}{2u}$

$\therefore \quad y' = y'$

$u'x + u = \frac{u}{2} - \frac{1}{2u}$

$u'x = \frac{u}{2} - u - \frac{1}{2u}$

$u'x = -\frac{u}{2} - \frac{1}{2u}$

$u'x = \frac{-u^2 - 1}{2u}$

$\frac{-2u^2 - 2}{2u}$

$\frac{du}{dx} \cdot x = \frac{-(1+u^2)}{2u}$

$\frac{2u \cdot du}{(1+u^2)} = -\frac{1}{x}\,dx$

$\frac{-2u}{(1+u^2)}\,du = \frac{1}{x}\,dx$ → Separable form.

Integrating ....

$\int \frac{-2u}{(1+u^2)}\,du = \int \frac{1}{x}\,dx + C \quad \text{—— ②}$

LHS;

$1+u^2 = t$

$2u \cdot du = dt$

$\therefore \int \frac{-2u}{(1+u^2)}\,du = \int \frac{-dt}{t}$

$= \int \frac{-dt}{t}$

$= -\ln t$

$= -\ln(1+u^2)$

② becomes.

$-\ln(1+u^2) = \ln x + C$

$-\ln(1+u^2) - \ln(x) = C$

$-\left[\ln(1+u^2) + \ln(x)\right] = C$

$cx - 1 = c^*$

$\ln\left(x(1+u^2)\right) = c^*$ → constant

$x(1+u^2) = e^{c^*} = k$ → another constant.

$u = \frac{y}{x}$

$x\left(1 + \left(\frac{y}{x}\right)^2\right) = k$

$x\left(1 + \left(y/x\right)^2\right) = k$

problems.

Q1: Solve $y \cdot y' + 36x = 0$

Q2: $xy' = y^2 + y$

Q3: $y' = 1 + 4y^2 \quad ; y(1) = 0$

1: $y \cdot y' + 36x = 0$

$yy' = -36x$

$y \cdot \frac{dy}{dx} = -36x$

$y \cdot dy = -36x \cdot dx$

$\int y \cdot dy = \int 36x\, dx + C$

$\frac{y^2}{2} = \frac{-36x^2}{2} + C$

$\frac{y^2}{2} = \frac{-36x^2}{2} + C$

$\frac{y^2}{2} + 36\frac{x^2}{2} = C$

$y^2 + 36x^2 = 2C = K$

Q:2: $xy' = y^2 + y$

$x \cdot \frac{dy}{dx} = y^2 + y$

$\frac{dy}{y^2 + y} = \frac{dx}{x}$

$\int \frac{1}{y^2+y} = \int \frac{1}{x} dx + C \rightarrow$ soln

LHS: $\int \frac{1}{y^2+y} = \int \frac{1}{y(y+1)}$ —

$\therefore$

$\frac{1}{y(y+1)} = \frac{A}{y} + \frac{B}{(y+1)}$

multiplying by $y(y+1)$;

$1 = \frac{A y(y+1)}{y} + \frac{B(y+1)(y)}{(y+1)}$

$1 = A(y+1) + By$

put $y = 0$

$1 = A(0+1) + B \times 0$

$A = 1$

put $y = -1$

$1 = A(-1+1) + B(-1)$

$1 = -B$

$B = -1$

$\therefore \frac{1}{y(y+1)} = \frac{1}{y} + \frac{-1}{(y+1)}$

$\frac{1}{y(y+1)} = \frac{1}{y} - \frac{1}{(y+1)}$

$\therefore$

LHS ;

$$\int \frac{1}{y(y+1)} = \int \left(\frac{1}{y} - \frac{1}{(y+1)}\right) dy$$

$$= \int \frac{1}{y} dy - \int \frac{1}{(y+1)} dy$$

$$= \ln y - \ln(y+1)$$

$$= \ln\left(\frac{y}{(y+1)}\right)$$

The soln ;

$$\ln\left(\frac{y}{(y+1)}\right) = \ln x + C$$

$$\ln\left(\frac{y}{y+1}\right) - \ln x = C$$

$\ln a - \ln b = c$
$\ln(a/b) = c$

$$\ln\left(\frac{y}{x(y+1)}\right) = C$$

$$\frac{y}{x(y+1)} = K$$

3. $y' = 1+4y^2$; $y(1) = 0$

$$y' = 1+4y^2$$

$$\frac{dy}{dx} = 1+4y^2$$

$$\frac{dy}{(1+4y^2)} = dx$$

$$\int \frac{1}{1+4y^2} \cdot dy = \int dx$$

$$= x$$

LHS ;

$$\int \frac{1}{1+4y^2} = \frac{\tan^{-1}(2y)}{2}$$

The soln

$$\frac{\tan^{-1}(2y)}{2} = x + C$$

$$\frac{\tan^{-1}(2y)}{2} - x = C$$

$\frac{A}{4}(y^2+1) \frac{B}{y^2} + \frac{B}{1/4}$

$\frac{4y^2+1}{y^2} + \frac{2^2}{1/4}$

$\int \frac{1}{4(y^2+1/4)}$

$\frac{1}{4} \int \frac{1}{y^2 \cdot \frac{1}{4}}$

$\frac{1}{4} \times 1$

Linear ODE : Bernoulli Eqn

A first order ODE

$$y' + p(x)y = r(x) \quad -(1)$$

where p, r are functions of x or constants.

This general eqn of (1) is

$$e^h y = \int e^h r(x)\,dx + C \quad -(2)$$

where, $h = \int p(x)\,dx$ called integrating factor.

Eqn (2) can be written as;

$$y = e^{-h}\left[\int e^h r(x)\,dx + C\right]$$

eg (1) Solve the initial value prblm,

$$y' + y\tan x = \sin 2x \;;\; y(0) = 1$$

ans: Since this is in Bernoulli Eqn;

$p(x) = \tan x$

$r(x) = \sin 2x$

Integrating factor;

$h = e^{\int p(x)\,dx}$

$h = \int p(x)\,dx$

$= \int \tan x\,dx$

$h = e^{\ln(\sec x)}$

$h = \sec x$

general soln;

$e^{\sec x} \cdot y = \int e^{\sec x} \sin 2x\,dx + C$

general soln;

$e^{\ln(\sec x)} y = \int e^{\ln \sec x} \cdot \sin 2x\,dx + C$ $\qquad e^{\ln \sec x} = \sec x$

$\sec x \cdot y = \int \sec x \cdot \sin 2x + C$

$\sec x \cdot y = \int \sec x \cdot 2\sin x \cdot \cos x + C$

$\sec x \cdot y = \int \frac{1}{\cos x} \cdot 2\sin x \cdot \cos x + C$

$\sec x \cdot y = \int 2\sin x + C$

$\sec x \cdot y = -2\cos x + C$

$y = -2\cos^2 x + C\cos x$

$y = -2\cos x - \sec x + C$

$y = -2\cos x - \frac{1}{\cos x} + C$

$C = 3$

$y = 2\cos^2 x$

$y = -2\cos$

$\therefore$ Soln;

$$y = 3\cos x - 2\cos^2 x$$

y. ②

$y' + 2y = 4\cos 2x$

$P(x) = 2$

$Q(x) = 4\cos 2x$.

$h = \int P(x)\,dx$

$= \int 2\,dx$

$= 2x$.

$e^h \cdot y = \int e^h \cdot Q(x) \cdot dx + C$

$e^{2x} \cdot y = \int e^{2x} \cdot 4\cos(2x)\,dx + C$

$\int uv = u\int v - \int\int v\,du$

$e^{2x} \cdot y = \int e^{2x} \cdot 4\cos(2x)\,dx + C$

RHS

$I = 4\int e^{2x} \cdot \cos 2x\,dx = e^{2x} \cdot \int 4\cos 2x - \int\left[4\cos 2x \cdot 2e^{2x}\,dx\right]$

$= e^{2x}\left(\frac{\sin 2x}{2}\right) - e^{2x}\int\left(\frac{\sin 2x}{2}\right) \cdot 2e^{2x}$

$= \frac{e^{2x} \cdot \sin 2x}{2} - \int e^{2x} \cdot \sin 2x \cdot dx$

$= \frac{e^{2x} \cdot \sin 2x}{2} - \left\{ e^{2x} \cdot \frac{-\cos 2x}{2} - \int \frac{-\cos 2x}{2} \cdot 2e^{2x}\,dx \right\}$

$I = \frac{e^{2x}\sin 2x}{2} + \frac{e^{2x}\cos 2x}{2} - \int e^{2x}\cos 2x\,dx$

$2I = \frac{e^{2x}\sin 2x + e^{2x}\cos 2x}{2}$

$I = \frac{e^{2x}}{4}(\sin 2x + \cos 2x)$

∴ gen soln:

$e^{2x} \cdot y = e^{2x}(\sin 2x + \cos 2x) + C$

$y = \sin 2x + \cos 2x + Ce^{-2x}$

③ $y' = 2y - 4x$

$y' - 2y = -4x$

$P(x) = -2$

$Q(x) = -4x$

$\int P(x)\,dx = \int -2\,dx = -2x$

$e^{-2x} \cdot y = \int e^{-2x} \cdot -4x \cdot dx + C$

$I = \int e^{-2x} \cdot -4x \cdot dx + 0$

$= \int -4x \cdot e^{-2x} + 0$

$= -4x \cdot e^{-2x} - \int\int e^{-2x} \cdot -4$

$= -4x \cdot e^{-2x} - \int -4e^{-2x} = -4x \cdot e^{-2x} - 4x \cdot e^{-2x}$

$\frac{d}{dx}(-4x) = -4$

$$= -4 \int x\, e^{-2x}\, dx + C$$

$$= -4 \left[x \cdot \frac{e^{-2x}}{-2} - \int \frac{e^{-2x}}{-2} \cdot 1\, dx \right] + C$$

$$= -4 \left\{ \frac{-x e^{-2x}}{2} + \frac{1}{2} \int e^{-2x}\, dx \right\} + C$$

$$e^{-2x} \cdot y = -2 \left\{ -x e^{-2x} + \frac{e^{-2x}}{2} \right\} + C$$

$\therefore$

$$y = 2x - 1 + C e^{2x}$$

(4) $y' + y \sin x = e^{\cos x}$

$P(x) = \sin x$

$Q(x) = e^{\cos x}$

$h = \int P(x)\, dx$

$= \int \sin x\, dx$

$= -\cos x$

$$e^{-\cos x} \cdot y = \int e^{-\cos x} \cdot e^{\cos x} \cdot dx + C$$

$$e^{-\cos x} \cdot y = x + C$$

$$y = e^{\cos x} x + e^{\cos x} C$$

Q: By modeling a first order ODE in R-L circuits, and hence find the current $I(t)$ in amperes, where t is the time. Assume that the circuit contain an EMF, $E(t) = 48$ volts of resistance of $R = 11\,\Omega$ & inductor of $L = 0.1$ Henrys. Initially the current is zero in the circuit.

PHYSICAL LAWS.

A current (I) in a circuit produces a voltage,

$V = IR$ (OHM's LAW)

and a voltage drop,

$LI' = L \frac{dI}{dt}$

Acc. to the Kirchoff's Law (KCL).

Emf produced = Sum of voltage drop & + voltage

$E(t) = RI + LI'$

$LI' + RI = E(t)$

$\div$ by L

$I' + \frac{R}{L} I = \frac{E(t)}{L}$ ↳ B.E form

$\therefore P(x) = \frac{R}{L} = \frac{11}{0.1}$

$Q(x) = \frac{E(t)}{L}$

$h = \int R/L\, dt = \frac{Rt}{L}$

The general soln ;

$$e^{\frac{Rt}{L}} I = \int e^{\frac{Rt}{L}} \cdot \frac{E(t)}{L} + C$$

$$\therefore I = e^{-\frac{Rt}{L}} \left\{ \frac{E}{L} \cdot \frac{e^{\frac{Rt}{L}}}{R/L} + C \right\}$$

$$I = \frac{E}{R} + C e^{-Rt/L}$$

Given ; $R = 11$, $L = 0.1$, $E = 48$

$$I(t) = \frac{48}{11} + C e^{-110t}$$

at $t = 0$ $I = 0$

$$I(0) = \frac{48}{11} + C e^{0} = 0$$

$$\therefore C = \frac{-48}{11}$$

∴ the current at 't'

$$I(t) = \frac{-48}{11} \left(1 - e^{-110t}\right)$$

UNIT-2 Linear Second Order ODE.

* A differential equation containing y'' or $\frac{d^2y}{dx^2}$ or y' or functions of x, y, and constants. is called a differential equations of second order.

The general form of a linear second order ODE is ;

★ $y'' + p(x)y' + q(x)y = r(x)$ ——①

if $r(x) = 0$, then ① is called Homogenous linear second order ODE, otherwise ① becomes non-homogenous

eg:- 1. $y'' + 25y = 0$ – Homogenous linear ODE of 2nd order.

2. $y'' + xy' + y = e^x$ – Non homogenous linear 2nd order ODE.

⇒ Fundamental theorem for homogenous linear ODE.

* For a homogenous linear ODE, any ~~two~~ linear combination of two solutions on an interval I is also a sol^n.

ie

if y_1, y_2 are solutions of $y'' + p(x)y' + q(x)y = 0$

then, $y = c_1y_1 + c_2y_2 \rightarrow c_1, c_2$ are arbitrary constants.

y is called the general solution of

$$y'' + p(x)y' + q(x)y = 0$$

eg:- 1 $y'' + y = 0$

Let $y_1 = \sin x$

$y_1' = \cos x$

$y_1'' = -\sin x$

$\therefore \quad y'' + y = 0$

$-\sin x + \sin x = 0 \quad \therefore$ is a soln, $y_1 = \sin x$

Let $y_2 = \cos x$

$y_2' = -\sin x$

$y_2'' = -\cos x$

$y_2'' + y_2 = 0$

$-\cos x + \cos x = 0 \quad \therefore \quad y_2 = \cos x$ is a soln

The general solution,

$$y = C_1 \sin x + C_2 \cos x$$

$$y' = C_1 \cos x - C_2 \sin x$$

$$y'' = -C_1 \sin x - C_2 \cos x$$

$$y'' + y = (-C_1 \sin x - C_2 \cos x) + (C_1 \sin x + C_2 \cos x)$$

$$= 0$$

$\therefore$ y is the general soln.

2] $y'' + y = 1$

$y_1 = 1 + \cos x \qquad y_2 = 1 + \sin x$

are solutions of ①

Let $y = y_1 + y_2$

$y = \cos x + \sin x + 2$

y is not a soln of ① ;

for, if $y' = -\sin x + \cos x$

$y'' = -\cos x - \sin x$

$y'' + y' = 2$

⇒ Linear second order ODE : IVP.

The general form of ODE.

$$y'' + p(x)y' + q(x)y = r(x) \; ; \; y(x_0) = k_0 \; ; \; y'(x_0) = k_1 \quad \text{---- ①}$$

after solving ①, we can determine the arbitrary constants in the general soln by applying the initial value condn

Then the soln become a particular soln of ①

eg:- $y'' + y = 0 \; ; \; y(0) = 3, \; y'(0) = -0.5$

The gen soln

$$y = C_1 \sin x + C_2 \cos x$$

Apply $y(0) = 3$

eg:- 1 $y'' + y = 0$

Let $y_1 = \sin x$

$y_1' = \cos x$

$y_1'' = -\sin x$

$\therefore \quad y'' + y = 0$

$-\sin x + \sin x = 0$ $\quad \therefore$ is a soln., $y_1 = \sin x$

Let $y_2 = \cos x$

$y_2' = -\sin x$

$y_2'' = -\cos x$

$y_2'' + y_2 = 0$

$-\cos x + \cos x = 0$ $\quad \therefore \quad y_2 = \cos x$ is a soln

The general solution,

$y = C_1 \sin x + C_2 \cos x$

$y' = C_1 \cos x - C_2 \sin x$

$y'' = -C_1 \sin x - C_2 \cos x$

$y'' + y = (-C_1 \sin x - C_2 \cos x) + (C_1 \sin x + C_2 \cos x)$

$= 0$

$\therefore$ y is the general soln.

2] $y'' + y = 1$

$y_1 = 1 + \cos x$ $\qquad$ $y_2 = 1 + \sin x$

are solutions of (1)

Let $y = y_1 + y_2$

$y = \cos x + \sin x + 2$

y is not a soln of (1);

for, $y' = -\sin x + \cos x$

$y'' = -\cos x - \sin x$

$y'' + y' = 2$

⇒ Linear second order ODE : IVP.

The general form of ODE.

$y'' + p(x)y' + q(x)y = r(x)$; $y(x_0) = k_0$; $y'(x_0) = k_1$ $\quad$ (1)

after solving (1), we can determine the arbitrary constants in the general soln by applying the initial value condn

Then the soln become a particular soln of (1)

eg:- $y'' + y = 0$; $y(0) = 3$, $y'(0) = -0.5$

The gen soln

$y = C_1 \sin x + C_2 \cos x$

Apply $y(0) = 3$

$3 = 0 + C_2$

$\therefore C_2 = 3$

$y' = C_1 \cos x - C_2 \sin x$

$y'(0) = -0.5$

$-0.5 = C_1 - 0$

$C_1 = -0.5$

∴ The particular soln

$y = -0.5 \sin x + 3 \cos x$

Basis

Let 'y_1' and 'y_2' are soln's of

$y'' + p(x)y' + q(x)y = 0$ ——(1)

if y_1, y_2 are independent, then

y_1, y_2 are called a basis of soln's of (1)

Note:- y_1, y_2 are independent if

$\frac{y_1}{y_2} \neq$ Constant

$y_1 = \sin x$, $y_2 = \cos x$

$\frac{y_1}{y_2} = \tan x$.

In the example,

$y_1 = \sin x$
$y_2 = \cos x$ are independent solutions of $y'' + y = 0$

hence $\{y_1, y_2\}$ form a basis of solution.

Q: Find a basis of soln $y'' - y = 0$

$y_1 = e^x$ $y_2 = e^{-x}$

$y_1' = e^x$ $y_2' = -e^{-x}$

$y_1'' = e^x$ $y_2'' = e^{-x}$

$y'' - y = 0$

$e^x - e^{-x} = 0$

clearly y_1 & y_2 are solutions

$\frac{y_1}{y_2} = \frac{e^x}{e^{-x}} = e^{2x} \neq$ constant

∴ y_1 & y_2 are independent.

hence it is a basis of soln

Reduction of Order.

Let $y'' + p(x)y' + q(x)y = 0$ — (1)

Let y_1 be a solution of (1).

we define $y_2 = uy_1$ — (2)

Let y_2 also be a solution, substituting y_2 in (1) and simplifying eqn (1) becomes.

$$u'' + u'\left(\frac{2y_1' + py_1}{y_1}\right) = 0 \quad \text{— (3)}$$

put $U = u'$.

(3) becomes

$$U' + U\left(\frac{2y_1' + py_1}{y_1}\right) = 0$$

by solving this equation,

$$U = \frac{1}{y_1^2} e^{-\int p(x)\,dx}$$

as $u' = U$

$$u = \int U\,dx$$

$$\therefore \quad y_2 = y_1 \int U\,dx$$

eg:- Solve the eqn $(x^2 - x)y'' - xy' + y = 0$ — (1).

$y_1 = x$ is a solution

$y_1 = x \qquad y_1'' = 0$

$y_2 = uy_1$

define $U = u'$

$$U = \frac{1}{y_1^2} e^{-\int p(x)\,dx}$$

eqn (1) in std form,

$$y'' - \left(\frac{x}{x^2 - x}\right)y' + \frac{y}{(x^2 - x)} = 0$$

$$p(x) = \frac{-x}{x^2 - x} = \frac{-1}{x-1}$$

$$U = \frac{1}{x^2} e^{-\int \frac{1}{x-1}\,dx}$$

$$= \frac{1}{x^2} e^{\ln(x-1)\,dx}$$

$$= \frac{x-1}{x^2}$$

$$u' = \frac{x-1}{x^2}$$

$$u = \int \frac{x}{x^2} - \frac{1}{x^2}$$

$$u = \int \frac{1}{x} - \frac{1}{x^2}$$

$u = \ln x + \frac{1}{x}$

$y = uy_1 = x(\ln x - \frac{1}{x})$

$y = x\ln x + 1$

general solution.

$y = c_1y_1 + c_2y_2$

$y = c_1x + c_2(x\ln x + 1)$

Linear Second Order ODE with Constant Coefficient.

The general form is :

$$y'' + ay' + by = r(x) \quad -①$$

Homogeneous $= y'' + ay' + by = 0 \quad -②$

To solve ②,

Let $y = e^{\lambda x}$ be a solution of ②

$y' = \lambda e^{\lambda x}$

$y'' = \lambda^2 e^{\lambda x}$

eqn ② becomes:

$\lambda^2 e^{\lambda x} + a\lambda e^{\lambda x} + be^{\lambda x} = 0$

$e^{\lambda x}(\lambda^2 + a\lambda + b) = 0$

$e^{\lambda x}$ can't be zero :

$\lambda^2 + a\lambda + b = 0$ → characteristic Eqn.

$\lambda = \frac{-a \pm \sqrt{a^2-4b}}{2}$

$\lambda_1 = \frac{-a + \sqrt{a^2-4b}}{2}$

$\lambda_2 = \frac{-a - \sqrt{a^2-4b}}{2}$

$y_1 = e^{\lambda_1 x}$ & $y_2 = e^{\lambda_2 x}$ are solutions of ②

CASE 1 (λ_1 and λ_2 are real & distinct)

In this case the two independent solutions of ② is given by

$y_1 = e^{\lambda_1 x}$ $\quad y_2 = e^{\lambda_2 x}$

$\left\{ \frac{y_1}{y_2} = e^{(\lambda_1 - \lambda_2)x} \neq \text{constant} \right\}$

the general solution is :

$$y = c_1e^{\lambda_1 x} + c_2e^{\lambda_2 x}$$

CASE 2 ($\lambda_1 = \lambda_2$)

$\lambda_1 = \lambda_2 = -\frac{a}{2}$

$y_1 = e^{-\frac{a}{2}x}$ $\quad y_2 = xe^{-\frac{a}{2}x}$

The general solution is :

$$y = c_1 e^{\frac{-ax}{2}} + c_2 x e^{\frac{-ax}{2}}$$

$$y = (c_1 + c_2 x) e^{-a/2 x}$$

CASE 3 (λ_1 & λ_2 are complex and imaginary.

$$\omega = \frac{\sqrt{a^2 - 4b}}{2}$$

let $\lambda_1 = \frac{-a}{2} + i\omega$ $\lambda_2 = \frac{-a}{2} - i\omega$

$y_1 = e^{\frac{-ax}{2}} \cos \omega x$, $y_2 = e^{\frac{-ax}{2}} \sin \omega x$

general solution :

$$y = e^{\frac{-ax}{2}} [A \cos \omega x + B \sin \omega x]$$

Problems.

Q1) Solve IVP.

$y'' + y' - 2y = 0$, $y(0) = 4$, $y'(0) = -5$

replacing y'' by λ^2 & y' by λ

$\lambda^2 + \lambda - 2 = 0$

$(\lambda + 2)(\lambda - 1) = 0$

$\lambda_1 = -2$ $\lambda_2 = 1$ {Case 1, real & distinct}

$y_1 = e^{-2x}$ $y_2 = e^{x}$

∴ general soln

$$y = c_1 e^{-2x} + c_2 e^{x} \quad \text{---(1)}$$

Apply $y(0) = 4$

(1) ⇒

$$4 = c_1 + c_2 \quad \text{---(2)}$$

now, $y' = -2c_1 e^{-2x} + c_2 e^{x}$

apply $y'(0) = -5$

$$-5 = -2c_1 e^{0} + c_2 e^{0}$$

$$-5 = -2c_1 + c_2 \quad \text{---(3)}$$

Solve (2) & (3)

$$c_1 + c_2 = 4$$
$$-2c_1 + c_2 = -5$$
$$3c_1 = 9$$

$c_1 = 3$

$c_2 = +1$

∴ The solution (1) ;

$y = e^{-2x} \cdot 3 + 1e^{x}$

$y = 3e^{-2x} + e^{x}$

Q: Solve $y'' + 6y' + 9y = 0$

Sum = 6
Pdt = 9

$$\lambda^2 + 6\lambda + 9 = 0$$

$$(\lambda+3)^2 = 0$$

$$\lambda + 3 = 0$$

$$\lambda_1 = \lambda_2 = -3$$

$\lambda_1 = \lambda_2 = 3$ {Case II, real & equal}

$y_1 = e^{-3x}$ $\quad y_2 = xe^{-3x}$

$$y = C_1 e^{-3x} + C_2 x e^{-3x}$$

$$y = e^{-3x}(C_1 + C_2 x) \quad \text{—①}$$

$y'' + y' + 0.25y = 0$, $y(0) = 3; y'(0) = -3.5$

Sum = 1
Pdt = $\frac{1}{4}$

$$\lambda^2 + \lambda + \frac{1}{4} = 0$$

$$(\lambda + \tfrac{1}{2})^2 = 0$$

$\lambda_1 = \lambda_2 = \frac{-1}{2} = -0.5$ {Case II, real & equal}

$y_1 = e^{-0.5x}$ $\quad y_2 = x \cdot e^{-0.5x}$

$$y = C_1 e^{-0.5x} + C_2 x \cdot e^{-0.5x}$$

$$y = e^{-0.5x}(C_1 + xC_2)$$

$y(0) = 3$

$3 = e^{-0.5(0)}(C_1 + 0C_2)$

$3 = C_1$

$y'(0) = -3.5$

$-3.5 = e^{-0.5\times 0}(0 + C_2)$

$C_2 = -3.5$

$y' = -0.5e^{-0.5x}(C_1 + x\times 0 + C_2 \times 1)$

$-3.5 = -0.5 \times e^0 (C_1 + C_2)$

$-3.5 = -0.5C_1 + -0.5C_2$

$0.5(3.5) = C_1 + C_2$

$-3.5 = -0.5\times 3. + -0.5C_2$

$3 = C_1 + C_2$

$-0.5C_2 = -3.5 + 1.5$

$C_2 = -3 - C_1$

$-0.5C_2 = -2$

$C_2 = -3 - 3$

-6

$C_2 = -2$

Q: Solve IVP. $y'' + 0.4y' + 9.04y = 0$, $y(0) = 0, y'(0) = 3$

$$\lambda^2 + 0.4\lambda + 9.04 = 0$$

$$\lambda = \frac{-0.4 \pm \sqrt{0.16 - 36.16}}{2}$$

$$= \frac{-0.4 \pm \sqrt{-36}}{2} = \frac{-0.4 \pm i6}{2}$$

$\lambda = -0.2 \pm 3i$

$-\alpha/2$; ω

general soln ;

$$y = e^{-0.2x}\left[A\cos 3x + B\sin 3x\right]$$

apply ; $y(0) = 0$

$$0 = e^{0}\left[A\cos(0) + B\sin(0)\right]$$

$$0 = A + 0$$

$$A = 0$$

$$y = e^{-0.2x} \cdot A\cos 3x + e^{-0.2x} \cdot B\sin 3x$$

$$y' = \left[e^{-0.2x} \cdot A \times 3\sin 3x + A\cos 3x \times -0.2e^{-0.2x}\right] + \left[e^{-0.2x} \cdot 3B\cdot\cos 3x + B\sin 3x \times -0.2e^{-0.2x}\right]$$

apply $y'(0) = 3$

$$3 = \left(e^{0} \times B \times \sin(0)\right) + \left(A \times 1 \times -0.2\right) + \left(e^{0} \times B \times 1\right) + \left(B \times 0 \times -0.2\right)$$

$$3 = -0.2A - 0.2B$$

$$3 = 3B - 0.2A$$

$$B = 1$$

3] $y'' + 25y = 0 \quad y(0) = 6.6 \quad y'(0) = -1.2$

$$\lambda^2 + 25 = 0$$

$$\lambda^2 = -25$$

$$\lambda = \sqrt{-25}$$

$$\lambda_1 = i5 \qquad \lambda_2 = -i5$$

$$\lambda = \pm i5$$

real part $= 0$

imaginary part $= 5$

general soln ;

$$y = e^{0}\left[A\cos 5x + B\sin 5x\right]$$

apply $y(0) = 6.6$

$$6.6 = A\cos(0) + B \times 0$$

$$A = 6.6$$

$$y = A\cos 5x + B\sin 5x$$

$$y' = -A \times 5\sin x + B \times 5\cos 5x$$

$-1.2 = 0 + 5B$

$5B = -1.2$

$B = -0.24$

Euler – Cauchy Equations.

The general form of an Euler-Cauchy equation is

$$x^2y'' + axy' + by = 0 \quad \text{—①}$$

assume that $y = x^m$ is a solution of ①

$\therefore \quad y' = mx^{m-1}$

$y'' = m(m-1)x^{m-2}$

Substituting in ①

$$x^2(m(m-1)x^{m-2}) + axmx^{m-1} + bx^m = 0$$

$$m(m-1)x^m + amx^m + bx^m = 0$$

$$x^m(m(m-1) + am + b) = 0$$

$$x^m(m^2 + (a-1)m + b) = 0$$

ie; Since $x^m \neq 0$

$$m^2 + (a-1)m + b = 0 \quad \text{—②} \rightarrow \text{Auxillary equation}$$

$m_1 = \frac{(1-a)}{2} + \sqrt{\frac{(1-a)^2}{4} - b}$, $m_2 = \frac{(1-a)}{2} - \sqrt{\frac{(1-a)^2}{4} - b}$

Case 1 m_1 & m_2 are real & distinct.

Independent solution.

$y_1 = x^{m_1} \qquad y_2 = x^{m_2} \qquad \left(\frac{y_1}{y_2} = x^{m_1 - m_2} \neq \text{constant}\right)$

$\therefore$ general solution,

$$y = C_1x^{m_1} + C_2x^{m_2}$$

Case 2 $m_1 = m_2 = m$ (real & equal)

$y_1 = x^m$

By using reduction of order, we can find the second Soln,

$y_2 = x^m \ln x$

$\therefore$ the general solution is

$$y = c_1x^m + c_2x^m \ln x$$

$$y = (c_1 + c_2 \ln x)x^m$$

Case 3 ~~m_1 & m_2 are complex roots.~~

Solve the Euler-Cauchy eqn:

$$x^2y'' - 5xy' + 9y = 0$$

Auxillary eqn ; $a = -5$, $b = 9$

$$m^2 + (-5-1)m + 9 = 0$$

$$m^2 - 6m + 9 = 0$$

$$m^2 - 6m + 9 = 0$$

$$(m-3)(m-3) = 0$$

$$m_1 = m_2 = 3$$

$$y_1 = x^3 \qquad y_2 = x^3 \ln x$$

∴ the general solution ;

$$y = C_1 x^3 + C_2 x^3 \ln x$$

$$y = x^3 (C_1 + C_2 \ln x)$$

Case 3 m_1 & m_2 are complex roots.

eg: $x^2 y'' + 0.6xy' + 16.04y = 0$

$a = 0.6$

$b = 16.04$

$$m^2 + (a-1)m + b = 0$$

$$m^2 + (0.6-1)m + 16.04 = 0$$

$$m^2 - 0.4m + 16.04 = 0$$ Auxillary Eqn.

$$m = \frac{0.4 \pm \sqrt{0.16 - 4 \times 16.04}}{2}$$

$$= \frac{0.4 \pm \sqrt{-64}}{2}$$

$$= \frac{0.4 \pm i8}{2}$$

$$= 0.2 \pm 4i$$

$m_1 = 0.2 + 4i$ $\qquad m_2 = 0.2 - 4i$

$$x^{m_1} = x^{0.2+4i} = x^{0.2} \cdot x^{4i} = x^{0.2} (e^{\ln x})^{4i} = x^{0.2} (\cos 4\ln x + i \sin 4\ln x)$$

$$x^{m_1} = x^{0.2} (\cos 4\ln x + i \sin 4\ln x)$$

$e^{\ln x} = x$

$\therefore (e^{\ln x})^{4i} = x^{4i}$

ie $e^{4i \ln x}$

$\Downarrow$

$e^{i4\ln x}$

$$x^{m_2} = x^{0.2} (\cos 4\ln x - i \sin 4\ln x)$$

Adding these two formulas and dividing the result by 2, we may obtain $x^{0.2}(\cos 4\ln x)$ is also a solution.

Subtracting eqns in I and dividing by $2i$, we may obtain $x^{0.2}(\sin 4\ln x)$ is also a soln.

∴ The independent solutions are ;

$$y_1 = x^{0.2} (\cos 4\ln x)$$

and

$$y_2 = x^{0.2} (\sin 4\ln x)$$

$e^{i\theta} = \cos\theta + i\sin\theta$

$e^{-i\theta} = \cos\theta - i\sin\theta$

∴ general soln;

$$y = x^{0.2}\left(A\cos(4\ln x) + B(\sin 4\ln x)\right)$$

In general, if $\alpha \pm i\beta$ are the roots,

$$y = x^{\alpha}(A\cos(\beta \ln x)$$

Examples:

1. $x^2y'' + 1.5xy' - 0.5y = 0$

$a = 1.5 \qquad b = -0.5$

$$m^2 + (a-1)m + b = 0$$

$$m^2 + (1.5-1)m - 0.5 = 0$$

$$m^2 + 0.5m - 0.5 = 0$$

$$(m - \tfrac{1}{2})(m + 1) = 0$$

$m_1 = \frac{1}{2} \qquad m_2 = -1$ ∴ Case 1 real & distinct.

$y_1 = x^{\frac{1}{2}} \qquad y_2 = x^{-1} = \frac{1}{x}$ (independent)

$y = C_1\sqrt{x} + \frac{C_2}{x}$ is the general solution.

2) $x^2y'' + 3xy' + y = 0 \; ; \; y(1) = 0 \; , y'(1) = 5$

$a = 3 \qquad b = 1$

AE ⇒

$$m^2 + (a-1)m + b = 0$$

$$m^2 + 2m + 1 = 0$$

$$(m+1)(m+1) = 0$$

$m_1 = m_2 = -1$ Case 2

$$y_2 = (C_1 + C_2 \ln x)x^{-1}$$

$y(1) = 0$

when $x = 1, y = 0$

∴ ~~$0 = C_1 + C_2 \ln 1 \times 1$~~

$$0 = (C_1 + C_2 \ln(1))\,1$$

$C_1 = 0$

$$y' = (C_1 + C_2 \ln x) - x^{-2} \times x^{-1} \times C_2 \times \frac{1}{x}$$

$y'(1) = 5$

$$5 = (C_1 + 0) \times -1 + C_2$$

$$5 = -C_1 + C_2$$

$C_2 = 5$

∴ the solution; $y = \frac{5\ln x}{x}$

$y' = C_1 \times 5x^4 + C_2 \times -3x^{-4}$

$y' = 5C_1 x^4 - 3C_2 x^{-4}$

$y'(1) = -4.5$

$-4.5 = 5C_1 - 3C_2 \quad -(2)$

$(1) \times 3$

$0.3 = 3C_1 + 3C_2 \quad -(3)$

$(3) + (4)$

$$\begin{array}{r} -4.5 = 5C_1 - 3C_2 \\ 0.3 = 3C_1 + 3C_2 \\ \hline -4.2 = 8C_1 \end{array} +$$

$C_1 = \frac{-4.2}{8}$

$= -0.525$

$C_1 + C_2 = 0.1$

$C_2 = 0.1 - C_1$

$= 0.1 + 0.525$

$= 0.625$

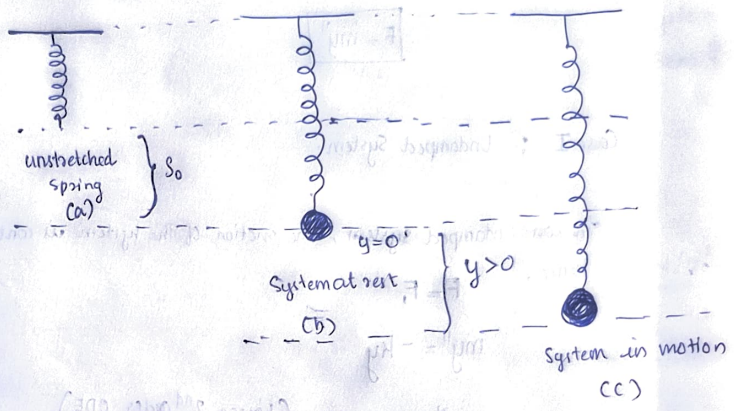
$y = C_1 x^5 + C_2 x^{-3}$

$y = -0.525 x^5 + 0.625 x^{-3}$

Modelling : Mass - Spring System.

Here we suspend an ordinary coil spring vertically from a fixed support and attatch a body at its lower end, say an iron ball.

Let $y=0$ denote the position of the ball when the system is at rest. we choose the downward dir. as +ve & upward as -ve



When the ball moves, the spring pull it down by an amnt $y>0$, this causes a spring force and is given by,

$$F_1 = -ky \quad \text{(Hooke's Law)}$$

where k is called "Spring constant"

Note: The spring is first unstretched (a) and then a weight is attatched (b) it causes a small displacement 's_0' due to a force 'F_0' acting in the upward direction. Hence by Hooke's law;

$$F_0 = -ks_0$$

But this force balances the weight of the body 'W' (∵ system at rest)

$F_0 + W = 0$

$W = -F_0$

$\boxed{W = ks_0}$

By Newton's IInd law :

mass x accelerat^n = Force

my'' = Force

$\boxed{F = my''}$

Case I :- Undamped System.

In an undamped system, the motion of the system is constant.

hence, $F = F_1$

$my'' = -ky$

$my'' + ky = 0$ (linear 2nd order ODE with constant coeff)

The characteristic Eqn:

$m\lambda^2 + k = 0$

$\lambda^2 = \frac{-k}{m}$ $\quad \omega_0 = \sqrt{k/m}$

$\therefore \lambda = \pm i\sqrt{k/m} = \pm i\omega_0$ (case 3)

∴ solution (motion)

$y(t) = A\cos\omega_0 t + B\sin\omega_0 t$ — (2)

"HARMONIC OSCILLATION".

Note :- The frequency of this motion, $\boxed{f = \frac{\omega_0}{2\pi}}$ Hz or cycles/sec

$\left\{\text{since period, } p = \frac{2\pi}{\omega_0}\right\}$

Note :- weight of body, $W = mg$, $g = 9.8\ m/s^2 = 32.17\ ft/sec^2$

eg:-1 Q. If a spring-mass system with an iron ball of weight $W = 98\ nt = 98\ N/m$ can be regarded as undamped and the spring is such that the ball stretches it 1.09 m. How many cycles per minute will the system execute. What will be the motion if we pull the ball from rest by 16cm & let it start with zero initial velocity.

$W = 98\ N/m$ $\qquad W = ks_0$

$s_0 = 1.09\ m$ $\qquad k = \frac{W}{s_0} = \frac{98}{1.09} = 89.9 \approx 90\ N/m^2$

$\approx 90\ kg/s^2$

$W = mg$

$m = \frac{98}{9.8} = 10\ kg$

$f = \frac{\omega_0}{2\pi} = \frac{\sqrt{k/m}}{2\pi} = \frac{\sqrt{90/10}}{2\pi} = \frac{3}{2\pi}\ Hz = 0.48\ Hz$

(1 Hz = 60 cycles/mt)

$f = 28.8$ cycles/mt

$f = 29$ cycles/mt

$y(t) = A\cos\omega_0 t + B\sin\omega_0 t$

$y(t) = A\cos 3t + B\sin 3t$

given $y(0) = 16cm = 0.16m$ $\quad y'(0) = 0$

$y(0) = 0.16$

$\therefore A\cos(0) + B\sin(0) = 0.16$

$A = 0.16$

$y' = -3A\sin 3t + 3B\cos 3t$

$y'(0) = 0$

$0 = -3A\sin(0) + 3B\cos(0)$

$0 = 3B$

$B = 0$

$\therefore y(t) = 0.16\cos 3t$

In a damped mass-spring system, their is a damping force proportional to rate of change of motion given by,

$$F_2 = -cy'$$

where c is damping constant.

In the previous model, we are adding this 'damping force, we would obtain

$$my'' = -ky - cy'$$

$$my'' + cy' + ky = 0$$

$$y'' + \frac{c}{m}y' + \frac{k}{m}y = 0 \longrightarrow (1)$$

The characteristic eqn of (1),

$$\lambda^2 + \frac{c}{m}\lambda + \frac{k}{m} = 0$$

The roots are,

$\lambda_1 = -\alpha + \beta$, $\lambda_2 = -\alpha - \beta$, where

$$\alpha = \frac{c}{2m} \quad \& \quad \beta = \frac{1}{2m}\sqrt{c^2 - 4mk}$$

Case (1) $c^2 > 4mk$.

when λ_1, λ_2 are distinct, real roots.

In this case, the general soln is,

$$y(t) = c_1 e^{(-\alpha+\beta)t} + c_2 e^{(-\alpha-\beta)t}$$

$$\boxed{y(t) = c_1 e^{(-\alpha-\beta)t} + c_2 e^{(-\alpha+\beta)t}}$$

the system has over damping. since, as $t \to \infty$, both terms in $y(t)$ tends to 0 $(y(t) \to 0)$

Than means, the system will be at rest whenever t is very large

Case (2) $c^2 = 4mk$

$\lambda_1 = \lambda_2 = -\alpha$, real & equal roots.

the general soln;

$$\boxed{y(t) = (c_1 + c_2 t)\, e^{-\alpha t}}$$

This case says that the system

Cas t becomes large, $y(t)$ slowly converges to zero (stable point)

Case ③ $c^2 < 4mk$.

Take $\beta = i\omega^*$, where $\omega^* = \frac{1}{2m}\sqrt{4mk - c^2}$

$$\omega^* = \sqrt{k/m - c^2/4m^2}$$

$\lambda_1 = -\alpha + i\omega^*$

$\lambda_2 = -\alpha - i\omega^*$.

general soln:

$$y(t) = e^{-\alpha t}(A\cos(\omega^* t) + B\sin(\omega^* t))$$

In this case, the system has can underdamping

Q:- How does the motion in eg1 change if we change the damping constant c with, $y(0) = 0.16$, $y'(0) = 0$

i) $c = 100$ kg/sec

ii) $c = 10$ kg/sec.

ans:- as in eg 10,

$m = 10$, $k = 90$, $c = 100$

The ODE is $(my'' + cy' + ky = 0)$

$10y'' + 100y' + 90y = 0$ ∴ $y(0) = 0.16$ $y'(0) = 0$

$y'' + 10y' + 9y = 0$

CE:

$\lambda^2 + 10\lambda + 9 = 0$

$(\lambda+1)(\lambda+9) = 0$

$\lambda_1 = -1$ $\lambda_2 = -9$ Case I (Overdamping)

Solution;

$y(t) = c_1 e^{-t} + c_2 e^{-9t}$

$y(0) = 0.16$

$0.16 = c_1 e^0 + c_2 e^0$

$0.16 = c_1 + c_2$ — ①

$y' = -c_1 e^{-t} - 9c_2 e^{-9t}$

$y'(0) = 0$

$0 = -c_1 - 9c_2$ — ②

① + ②

$-8c_2 = 0.16$

$c_2 = -0.02$

$c_1 = 0.18$

The motion at time 't'

$y(t) = 0.18e^{-t} - 0.02e^{-9t}$

2) $C = 60$, $m = 10$, $k = 90$

$my'' + cy' + ky = 0$

$10y'' + 60y' + 90y = 0$

$y'' + 6y' + 9y = 0$

C.E $\lambda^2 + 6\lambda + 9 = 0$

$(\lambda+3)(\lambda+3) = 0$

$\lambda_1 = \lambda_2 = -3$ Case II (Critical damping)

$y = (c_1 + c_2 t)e^{-\lambda t}$

$y(t) = (c_1 + c_2 t)e^{-3t}$

$y(0) = 0.16$

$0.16 = (c_1 + 0)e^0$

$0.16 = c_1$

$c_1 = 0.16$

$y' = -3(c_1 + c_2 t)e^{-3t} + e^{-3t} \times c_2$

$y' = -3(c_1 + c_2 t)e^{-3t} + c_2 e^{-3t}$

$y'(0) = 0$

$0 = -3(c_1 + 0) \times 1 + c_2 \times 1$

$0 = -3c_1 + c_2$

$c_2 = 3c_1 = 0.48$

$c_2 = 0.48$

$y = (0.16 + 0.48t)e^{-3t}$

3) $C = 10$

The ODE;

$10y'' + 10y' + 90y = 0$ $y'(0) = 0$, $y(0) = 0.16$

$y'' + y' + 9y = 0$

$\lambda = \frac{-1 \pm \sqrt{1^2 - 4 \times 1 \times 9}}{2} = \frac{-1 \pm \sqrt{1-36}}{2} = \frac{-1 \pm \sqrt{-35}}{2}$

$\lambda = -0.5 \pm 2.96i$

$\lambda_1 = -0.5 + 2.96i$ $\lambda_2 = -0.5 - 2.96i$

∴ gen eqn;

$y(t) = e^{-0.5t}(A \cos 2.96t + B \sin 2.96t)$

$y(0) = 0.16$

$0.16 = 1\,(A \cos 0 + B \sin 0)$

$0.16 = A$

$y' = e^{-0.5t}(-2.96A\sin(2.96t) + 2.96B\cos(2.96t))$

$y'(0) = 0$

$0 = 2.96B$

$B = 0$

$$y(t) = 0.16e^{-0.5t}\cos(2.96t)$$

Non-Homogenous Second ORDER Liner ODE

The general form is

$$y'' + p(x)y' + q(x)y = r(x) \quad -(1)$$

where p, q, r are functions of x.

The general form of non-homogeneous Linear Second order ODE with constant coefficients is given by,

$$y'' + py' + qy = r(x) \quad -(2)$$

$p, q \Rightarrow$ constants

$r \Rightarrow$ function of x.

11/02/26 Tuesday

Solution

The general soln of (1) is in the form:

$$y = y_h(x) + y_p(x) \quad -(3)$$

where $y_h(x)$ is the general solution of the corresponding homogenous eqn. and $y_p(x)$ is any particular soln of eqn (1)

eg:- $\Rightarrow$

$y'' + 2y' + 3y = \cos x$

I Method of undetermined coefficients (To find $y_p(x)$).

We can find out a particular solution $y_p(x)$ based on the nature the RHS function, $r(x)$

The following table gives the choice of y_p w.r.t $r(x)$

Term in $r(x)$	choice for $y_p(x)$
Ke^{ax}	Ce^{ax}
Kx^n $(n=0,1,\dots)$	$K_0 + K_1x + K_2x^2 + \dots + K_nx^n$
$K\cos\omega x$ $K\sin\omega x$	$K\cos\omega x + M\sin\omega x$
$Ke^{ax}\cos\omega x$ $Ke^{ax}\sin\omega x$	$e^{ax}(K\cos\omega x + M\sin\omega x)$

CHOICE RULES. FOR y_p

a) Basic Rule.

if $r(x)$ is one of the f^n in the first column in the table, then choose y_p in the same line in the table and determine its undetermined co-effs by substituting y_p and its derivatives in (1).

b) Modification Rule.

If a term in y_p happens to be a solution of the corresponding homogeneous ODE. Then multiply this term by x (or by x^2, if the soln becomes a double root of homogeneous ODE)

c) Sum rule.

If $r(x)$ is the sum of the functions in the first column of the table, then choose y_p as the sum of the functions in the corresponding lines of the second column of the table.

eg: $r(x) = xe^{2x}$

$y_p = K_0 + K_1 x + Ce^{2x}$

EXAMPLES.

1. Solve $y'' + y = 0.001x^2$, $y(0) = 0$, $y'(0) = 1.5$

$$\boxed{y = y_h + y_p}$$

to find y_h ...

H.E $y'' + y = 0$

C.E $\lambda^2 + \lambda = 0$

$\lambda = \pm i$ (case 3)

$y_h = A\cos x + B\sin x$.

now, $r(x) = 0.001x^2$

choose;

$y_p = K_0 + K_1 x + K_2 x^2$ as a particular soln

$y_p' = K_1 + 2K_2 x$

$y_p'' = 2K_2$

Substituting in eqn $(y'' + y = 0.001x^2)$

$\therefore$ $y_p'' + y_p = 0.001x^2$

$2K_2 + K_0 + K_1 x + K_2 x^2 = 0.001x^2$

equating the coefficients of x on both sides.

$2K_2 + K_0 = 0$

$K_1 = 0$

$K_2 = 0.001$

$\therefore K_0 = -2 \times K_2$

$K_0 = -0.002$

$\therefore y_p = -0.002 + 0.001x^2$

$\therefore$ the general solution is;

$y = y_h + y_p$

$y = A\cos x + B\sin x - 0.002 + 0.001x^2$

Given, $y(0) = 0$

ie,

$0 = A \times 1 + B \times 0 - 0.002 + 0$

$0 = A - 0.002$

$A = 0.002$

given; $y'(0) = 1.5$

$y' = -A\sin x + B\cos x + 0.002x$

$y'(0) = 1.5$

$1.5 = -A \times 0 + B \times 1 + 0$

$B = 1.5$

$\therefore y = 0.002\cos x + 1.5\sin x + 0.001x^2 - 0.002$.

2. $y'' + 3y' + 2.25y = -10e^{-1.5x}$ $y(0) = 1$, $y'(0) = 0$

to solve y_h

$y'' + 3y' + 2.25 = 0$

$\lambda^2 + 3\lambda + 2.25 = 0$

$(\lambda + 1.5)(\lambda + 1.5) = 0$

$\lambda_1 = \lambda_2 = -1.5$ (double root).

$y_h = (C_1 + C_2 x)e^{-\alpha x}$

$\therefore y_h = (C_1 + C_2 x)e^{-1.5x}$

to find y_p.

$r(x) = -10e^{-1.5x}$

$\times x^2 \ldots$ (modification rule)

choose $y_p = Cx^2e^{-1.5x}$ $\{\because e^{-1.5x}$ is a soln of Homogeneous ODE w.r.t double root $-1.5\}$

as the partial solution.

$y_p' = C\left[x^2 \cdot -1.5 \cdot e^{-1.5x} + e^{-1.5x} \cdot 2x\right]$

$y_p' = C\left[-1.5x^2e^{-1.5x} + 2xe^{-1.5x}\right]$

$y_p' = Ce^{-1.5x}\left[2x - 1.5x^2\right]$

$y_p'' = C\left[e^{-1.5x}(2 - 3x) + (2x - 1.5x^2) \times -1.5e^{-1.5x}\right]$

$y_p'' = Ce^{-1.5x}\left[2 - 3x - 3x + 2.25x^2\right]$

Substitute y_p, y_p', & y_p'' in the eqn;

$Ce^{-1.5x}\left(2 - 6x + 2.25x^2\right) + 3Ce^{-1.5x}(2x - 1.5x^2) + 2.25Cx^2e^{-1.5x} = -10e^{-1.5x}$

Equating the constant term on both sides; (without x or x^2 or....)

$2C = -10$

$C = -5$

$y_p = Cx^2 e^{-1.5x}$

$y_p = -5x^2 e^{-1.5x}$

∴ general solution, $y = y_h + y_p$

$y = (c_1 + c_2 x) e^{-1.5x} - 5x^2 e^{-1.5x}$

given $y(0) = 1$

$1 = (c_1 + 0) e^0 - 5 \times 0 \times e^0$

$1 = c_1$

$c_1 = 1$

given $y'(0) = 0$

$y' = \left[(c_1 + c_2 x) \times -1.5 e^{-1.5x} + e^{-1.5x} \times c_2\right] - 5\left[x^2 \times -1.5 e^{-1.5x} + e^{-1.5x} \cdot 2x\right]$

$y'(0) = 0$

$0 = (c_1 + 0) \times -1.5 \times 1 + 1 \times c_2 - 5[0 + 0]$

$0 = -1.5 c_1 + c_2$

$c_2 = 1.5$

$y = (1 + 1.5x) e^{-1.5x} - 5x^2 e^{-1.5x}$

3) Solve $y'' - 3y' - 2y = 10 \sin x$, $y(0) = 1$, $y'(0) = 6$

To find y_h

$y'' - 3y' - 2y = 0$

$\lambda^2 - 3\lambda - 2 = 0$

$(\lambda - 1)(\lambda - 2) = 0$

$\lambda_1 = 1 \quad \lambda_2 = 2$ (Case I)

$y_h = C_1 e^x + C_2 e^{2x}$

To find y_p

Here, $r(x) = 10 \sin x$

choose, $y_p = K \cos x + M \sin x$ — a particular Sol^n

$y_p' = -K \sin x + M \cos x$

$y_p'' = -K \cos x - M \sin x$

∴ Eqn become

$-K \cos x - M \sin x - 3(-K \sin x + M \cos x) + 2(K \cos x + M \sin x) = 10 \sin x$

$(-K - 3M + 2K) \cos x + (-M + 3K + 2M) \sin x = 10 \sin x$

$(-3M + K) \cos x + (M + 3K) \sin x = 10 \sin x$

Equating the coeff

$-3M + K = 0$ — ①

$M + 3K = 10$ — ②

① × 3

$-9M + 3K = 0$

$M + 3K = 10$

$-10M = -10$

$M = 1$

from ①;

$-3 + K = 0$

$K = 3$

$\therefore y_p = 3\cos x + \sin x$

∴ general soln

$y = y_h + y_p$

$y = c_1 e^x + c_2 e^{2x} + 3\cos x + \sin x$

given $y(0) = 1$

$1 = c_1 + c_2 + 3$

$c_1 + c_2 = -2$ — ③

~~diff~~ $y' = c_1 e^x + 2c_2 e^{2x} - 3\sin x + \cos x$

given $y(0) = 6$

$6 = c_1 + 2c_2 + 1$

$c_1 + 2c_2 = 5$ — ⑤

⑤ − ③;

$c_1 + 2c_2 = 5$

$c_1 + c_2 = -2$

$c_2 = 7$

$c_1 = -2 - c_2 = -9$

$c_1 = -9$

Soln;

$y = -9e^x + 7e^{2x} + 3\cos x + \sin x$

4) $y'' + 4y' + 4y = e^{-x}\cos x$.

$y'' + 4y' + 4y = 0$

$m^2 + 4m + 4 = 0$

$(m+2)(m+2) = 0$

$m_1 = m_2 = -2$ (Case II)

$y_h = (c_1 + c_2 x) e^{-2x}$

$g(x) = e^{-x}\cos x$

$\therefore$ $y_p = e^{-x}(K\cos x + M\sin x)$

$y_p' = e^{-x}(-K\sin x + M\cos x) + (K\cos x + M\sin x)\cdot -1e^{-x}$

$y_p' = e^{-x}(-K\sin x + M\cos x) - e^{-x}(K\cos x + M\sin x)$

$* \; y_p' = e^{-x}((M-K)\cos x + (-K-M)\sin x)$

$y_p'' = \left[e^{-x}(-K\cos x - M\sin x) - e^{-x}(-K\sin x + M\cos x)\right] - \left[e^{-x}(-K\sin x + \right.$

$\left. -e^{-x}(K\cos x \right.$

$y_p'' = \left[e^{-x}\left[(-K+K-M-M)\cos x + (-M+M+M+K)\sin x\right]\right.$

$* \; y_p'' = e^{-x}\left[2K\sin x - 2M\cos x\right]$

$\therefore$ Eqn becomes;

$e^{-x}(2K\sin x - 2M\cos x) + 4e^{-x}((M-K)\cos x + (-M-K)\sin x) +$

$4e^{-x}(K\cos x + M\sin x) = e^{-x}\cos x$

$2K - 4M - 4K + 4M = 0$

$-2K = 0$

$K = 0$

$-2M + 4M - 4K + 4K = 1$

$2M = 1$

$M = 1/2$

$\therefore \; y_p = e^{-x}(0\times\cos x + \frac{1}{2}\sin x)$

$y_p = e^{-x}\,\frac{1}{2}\sin x$.

$\therefore$ gen soln;

$y_h + y_p$;

$= (C_1 + C_2 x)e^{-2} + \frac{1}{2}e^{-x}\sin x$.

5) Solve $y'' + 2y' + 0.75y = 2\cos x - 0.25\sin x + 0.09x$

$y(0) = 2.78$, $y'(0) = -0.43$.

$y'' + 2y' + 0.75y = 0$ $\qquad * \; y_h = C_1e^{-x/2} + C_2e^{-3x/2}$

$\lambda^2 + 2\lambda + 0.75 = 0$

$(\lambda + \frac{1}{2})(\lambda + \frac{3}{2}) = 0$

$\lambda_1 = -\frac{1}{2} \quad \lambda_2 = -\frac{3}{2}$ Case I

$y_h = C_1e^{-x/2} + C_2e^{-3x/2}$

To find y_p

Let $y_p = y_{p_1} + y_{p_2}$

$y_{p_1} = K\cos x + M\sin x$

$y_{p_2} = K_0 + K_1 x$

$y_{p_1}' = -K\sin x + M\cos x$

$y_{p_1}'' = -K\cos x - M\sin x$

$y_{p_2}' = K_1$

$y_{p_2}'' = 0$

$y_{p_1}' = -K\sin x + M\cos x$.

$y_{p_1}(0) = 2.78$

$K = 2.78$

$y_{p_1} = K\cos x + M\sin x$

$y_{p_1}(0) = 2.78$

$K = 2.78$

$y_{p_1}'' = -K\cos x - M\sin x$

$y_{p_1}''(0) = -0.43$

$-K = -0.43$

$K = 0.43$

$\therefore y_{p_1} = 0.43\cos x + 0.78\sin x$

$y_{p_2}' = K_1$

$y_{p_1}'(0) = -0.43$

$-0.43 = M$

$M = -0.43$

$y_{p_1} = 2.78\cos x - 0.43\sin x$.

$y_{p_2} = K_0 + K_1 x$

$y_{p_2}(0) = 2.78$

$K_0 = 2.78$

$y_{p_2}' = K_1$

$y_{p_2}'(0) = -0.43$

$K_1 = -0.43$

$y_{p_2} = 2.78 - 0.43x$.

$\therefore$ gen soln.

$y = y_b + y_p$

$y = C_1 e^{-x/2} + C_2 e^{-3x/2} + 2.78\cos x - 0.43\sin x + 2.78 - 0.43x$

$y(0) = 2.78$ $\therefore$

$2.78 = C_1 + C_2 + 2.78 - 0 + 2.78 - 0$

$-2.78 = C_1 + C_2$

$C_1 + C_2 = -2.78$ —①

$y' = -C_1 \frac{1}{2} e^{-x/2} + \frac{3}{2} C_2 e^{-3x/2} = 2.78 \sin x - 0.43 \cos x - 0.4$

$y'(0) = -0.43$

$-0.43 = -\frac{C_1}{2} - \frac{3}{2} C_2 - 0 - 0.43 - 0.43$

$0.43 = -\frac{C_1}{2} - \frac{3C_2}{2}$ —②

② × 2

$0.86 = -C_1 - 3C_2$

$-2.78 = C_1 + C_2$ +

$-1.92 = 4C_2$

$C_2 = \frac{-1.92}{4}$

$C_2 = -0.48$

$C_1 = -2.78 - (-0.48)$

$C_1 = -2.3$

$\therefore y = -2.3 e^{-x/2} - 0.48 e^{-3x/2} + 2.78 \cos x - 0.43 \sin x - 0.43x$

Q: Solve the $y'' + 2y' + 0.75y = 2\cos x - 0.25\sin x + 0.09x$

$y(0) = 2.78$, $y'(0) = -0.43$

$y_h = C_1 e^{-x/2} + C_2 e^{-3x/2}$

To find y_p :

Let $y_p = y_{p_1} + y_{p_2}$ (Separating and choices)

$y_{p_1} = K\cos x + M\sin x$

$y_{p_2} = K_0 + K_1 x$

$y'_{p_1} = -K\sin x + M\cos x$

$y''_{p_1} = -K\cos x - M\sin x$

$y'' + 2y' + 0.75y = 2\cos x - 0.25\sin x + 0.09x$

$-K\cos x - M\sin x + 2(-K\sin x + M\cos x) + 0.75(K\cos x + M\sin x)$

$= 2\cos x - 0.25\sin x + 0.09x$

Equate the coeff of cos and sin x ,

$(-K + 2M + 0.75K)\cos x + (-M - 2K + 0.75M)\sin x = 2\cos x - 0.25\sin x + 0.09x$

$-0.25K + 2M = 2$ —①

$-2K + (-0.25M) = -0.25$ —②

Solving ① & ②

$K = 0$

$M = 1$

$y_{p_2} = K_0 + K_1 x$

$y'_{p_2} = K_1$

$y''_{p_2} = 0$

Eqn becomes ;

$0 + 2K_1 + 0.75(K_0 + K_1 x) = 2\cos x - 0.25\sin x + 0.09x$

Equating coeff of x^2 & x .

$2K_1 + 0.75K_0 = 0$

$0.75K_1 = 0.09$

$K_1 = 0.12$

$\therefore K_0 = -0.32$

$\therefore y_p = \sin x - 0.32 + 0.12x$

$\therefore$ General Solution, $y = C_1 e^{-x/2} + C_2 e^{-3x/2} + \sin x - 0.32 + 0.12x$

$y(0) = 2.78$.

$2.78 = C_1 + C_2 - 0.32$

$\therefore C_1 + C_2 = 3.1$ —①

$y' = -\frac{1}{2} C_1 e^{-x/2} - \frac{3}{2} C_2 e^{-3x/2} + \cos x + 0.12$

$y'(0) = -0.43$

$-0.43 = \frac{-C_1}{2} - \frac{3C_2}{2} + 1$

$-0.86 = -C_1 - 3C_2 + 1.12$

$\therefore C_1 + 3C_2 = 1.98$ —②

Solving ① and ②

$C_1 = 3.1$ $\quad C_2 = 0$

$\therefore$ Solution ; $y = 3.1 e^{-x/2} + \sin x + 0.12x - 0.32$

WRONSKIAN.

Let $y_1, y_2, \cdots, y_n$ are n functions of x, defined on an interval I and n^{th} order derivatives exist, then the wronskian is defined as :

$$W(y_1, y_2, \cdots, y_n) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ y_1'' & y_2'' & \cdots & y_n'' \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}$$

Theorem 1

Let $y_1, y_2, \cdots y_n$ be functions of x defined on an interval I, they are linearly independent on I, iff (if and only if), $W \neq 0$ for all $x \in I$.

if $W = 0$ for at least one x in I, then $y_1, y_2, \cdots y_n$ are linearly dependent.

For eg :- $y_1 = \cos x$, $y_2 = \sin x$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$= \cos^2 x + \sin^2 x$

$= 1 \neq 0$

$\therefore y_1, y_2$ are independent //

IInd Method.

Solution by Method of Variation of Parameters.

Consider a second order linear ODE, $y'' + py' + qy = r(x)$ —— (1)

The given soln (1) is :,

$$y = y_h + y_p$$

where y_h is the general soln of the corresponding homogenous ODE. Let the general soln, $y_h = c_1 y_1 + c_2 y_2$

(y_1, y_2 are independent soln, of Homogenous ODE)

By lagrange's method, the particular solution :

$$y_p(x) = -y_1 \int \frac{y_2 r(x)}{W} dx + y_2 \int \frac{y_1 r(x)}{W} dx$$

(W → wronskian value.)

eg:- (1) Solve $y'' + y = \sec x$

$$y = y_h + y_p$$

To find y_h

$y'' + y = 0$

C.E : $\lambda^2 + 1 = 0$

$\lambda = \pm i$ Case (III)

The Basis of Solutions are :

$y_1 = \cos x$

$y_2 = \sin x$

$y_h = A\cos x + B\sin x$.

$$y_p = -y_1 \int \frac{y_2 r}{W} dx + y_2 \int \frac{y_1 r}{W} dx$$

Here, $\omega = 1$

$$\int \frac{y_2 r}{W} dx = \int \frac{\sin x \sec x}{1} dx = \int \frac{\sin x}{\cos x} dx = \int \tan x \, dx$$

$$= \ln(\sec x)$$

$$\int \frac{y_1 r}{W} dx = \int \frac{\cos x \cdot \sec x}{1} dx = \int 1 \, dx = x$$

$y_p = -\cos x \ln(\sec x) + \sin x \cdot x \Rightarrow \cos x \ln(\cos x) + x \sin x$

$\therefore$ Gen soln :

$y = y_h + y_p$

$= A\cos x + B\sin x + (-\cos x \ln(\sec x)) + x\sin x$

$y = A\cos x + B\sin x - \cos x \ln(\sec x) + x \sin x$.

ve by method of variation of parameters.

2.

Soln:

$x^2y'' - 2xy' + 2y = x^3 \sin x$ — (1)

$y = y_h + y_p$

To find y_h ;

H-E ; $x^2y'' - 2xy' + 2y = 0$ (Euler - Cauchy eqn)

Auxillary Eqn ;

$m^2 + (a-1)m + b = 0$

$a = -2$, $b = 2$

$m^2 - 3m + 2 = 0$

$(m-1)(m-2) = 0$

$m = 1, 2$ Case I

$y_1 = x$, $y_2 = x^2$

$y_h = c_1x + c_2x^2$.

$y_p(x) = -y_1 \int \frac{y_2 r}{W} dx + y_2 \int \frac{y_1 r}{W} dx$

$W(y_1, y_2) = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = 2x^2 - x^2 = x^2$

general form of (1) ($\div$ by x^2)

$y'' - \frac{2}{x}y' + \frac{2}{x^2}y = x \sin x$.

$r(x) = x \sin x$ //

$\int \frac{y_2 r}{W} dx = \int \frac{x^2 \cdot x \sin x}{x^2} \cdot dx$

$= \int x \sin x \, dx$

$= x \times -\cos x - \int -\cos x \times 1 \cdot dx$

$= -x \cos x + \sin x$.

$\int \frac{y_1 r}{W} dx = \int \frac{x \cdot x \sin x}{x^2} \cdot dx = -\cos x$

The particular solution is ;

$y_p = -x(-x\cos x + \sin x) + x^2 \times -\cos x$

$= x^2 \cos x - x \sin x - x^2 \cos x$

$y_p = -x \sin x$

The general solution is ;

$y = c_1x + c_2x^2 - x\sin x$.

3) $(D^2 - 4D + 4I)y = \frac{6e^{2x}}{x^4}$ where $Dy = \frac{dy}{dx} = y'$

$D^2y = \frac{d^2y}{dx^2} = y''$

$y'' - 4y' + 4y = \frac{6e^{2x}}{x^4}$

$y = y_h + y_p$

To find y_h ;

$y'' - 4y' - 4y = 0$

C.E $\quad \lambda^2 - 4\lambda - 4 = 0$

$(\lambda - 2)(\lambda - 2) = 0$

$\lambda = 2, 2$ Case II

$y_1 = e^{2x} \qquad y_2 = xe^{2x}$

$y_h = (c_1 + c_2 x)e^{2x}$

① is in general form;

$r(x) = \frac{6e^{2x}}{x^4}$

$y_p = -y_1 \int \frac{y_2 r}{W} dx + y_2 \int \frac{y_1 r}{W} dx$

$W(y_1, y_2) = \begin{vmatrix} e^{2x} & xe^{2x} \\ 2e^{2x} & 2xe^{2x} + e^{2x} \end{vmatrix}$

$W = 2xe^{4x} + e^{4x} - 2xe^{4x}$

$W = e^{4x}$

Now;

$$\int \frac{y_2 r}{W} dx = \int \frac{xe^{2x} \cdot \frac{6e^{2x}}{x^4}}{e^{4x}} dx$$

$$= 6 \int \frac{1}{x^3} dx$$

$$= 6 \times \frac{x^{-2}}{-2} = -3x^{-2}$$

$$\int \frac{y_1 r}{W} dx = \int \frac{e^{2x} \frac{6e^{2x}}{x^4}}{e^{4x}} dx$$

$$= 6 \int \frac{1}{x^4} dx$$

$$= 6 \times \frac{x^{-3}}{-3}$$

$$= -2x^{-3}$$

The particular soln;

$$y_p = -e^{2x} \cdot 3x^{-2} + xe^{2x} \cdot -2x^{-3}$$

$$= 3x^{-2}e^{2x} - 2e^{2x}x^{-2}$$

$$= \frac{e^{2x}}{x^2}$$

The general soln;

$$y = (c_1 + c_2 x)e^{2x} + \frac{e^{2x}}{x^2} //$$

4) $(x^2D^2 + xD - 9I)y = 48x^5$

$x^2y'' + xy' - 9y = 48x^5$

$m^2 + (a-1)m + b = 0$

$a = 1 \quad b = -9$

$m^2 - 9 = 0$

$m^2 = 9$

$m = \pm 3$

$y_1 = x^{-3} \qquad y_2 = x^3$

$y_h = C_1 x^{-3} + C_2 x^3$

$$W(y_1, y_2) = \begin{vmatrix} x^{-3} & x^3 \\ -3x^{-4} & 3x^2 \end{vmatrix} = 3x^{-1} + 3x^{-1}$$

$$= 6x^{-1}$$

$$= 6/x$$

$$y_p = -y_1 \int \frac{y_2 r}{W} dx + y_2 \int \frac{y_1 r}{W} dx$$

$$\int \frac{y_2 r}{W} dx = \int \frac{x^3 \cdot 48x^3}{6/x} dx$$

$$= 8 \int x^7 dx$$

$$= 8 \times \frac{x^8}{8} = x^8$$

$$\int \frac{y_1 r}{W} dx = \int \frac{x^{-3} \cdot 48 \cdot x^{3+1}}{6/x} dx$$

$$= \int \frac{48x}{6}$$

$$= \frac{8x^2}{2}$$

$$= 4x^2$$

The particular soln;

$$y_p = -x^{-3} \times x^8 + x^3 \times 4x^2$$

$$= -x^5 + 4x^5$$

$$y_p = 3x^5$$

∴ the general soln;

$$y = C_1 x^{-3} + C_2 x^3 + 3x^5$$

Tutorial I [First Order ODE]

* $y' = f(x, y)$

* exponential decay prblm;

$y' = -ky \quad k > 0$

* exponential growth prblm;

$y' = ky \quad k$

$y(t) = y_0 e^{-kt}$

* non-separable form;

eg: $\frac{dy}{dx} = \frac{x+y}{x}$

$y' = \frac{x+y}{x} = 1 + \frac{y}{x}$

put $y = vx$ $\quad v = \frac{y}{x}$

$y' = v + v'x$

$\therefore v + v'x = 1 + v$

$v'x = 1$

$\frac{dv}{dx} \cdot x = 1$

$\int dv = \int \frac{1}{x} dx$

$v = \ln x + C$

$\frac{y}{x} = \ln x + C$

1. A substance initially has 500g. After 3hrs it reduces to 400g. Find the decay constant and the amnt after 8 hrs.

$y_0 = 500 \qquad y' = -ky$

$y(3) = 400$

$y(t) = y_0 e^{-kt}$

$400 = 500 \times e^{-3k}$

$e^{-3k} = \frac{400}{500}$

$e^{-3k} = \frac{4}{5}$

$-3k = \ln(4/5)$

$k = \frac{-1}{3} \ln(4/5)$

$k = \frac{1}{3} \ln(5/4)$

$k = 0.074$

$y(8) = y_0 e^{-0.074 \times 8}$

$= 500 \times e^{-0.074 \times 8}$

$= 376.6 \text{ g}$

2) i) $\frac{dy}{dx} = \frac{x}{1+y^2}$ $\quad y(0)=1$

$\int (1+y^2)\,dy = \int x\,dx$

$y + \frac{y^3}{3} = \frac{x^2}{2} + C$

$y = \frac{x^2}{2} - \frac{y^3}{3} + C$

ii) $\frac{dy}{dx} = \frac{y(1+x^2)}{x(1+y^2)}$

$\int \frac{(1+y^2)}{y}dy = dx\,(\frac{1+x^2}{x})$

$\int \frac{1}{y} + y\,dy = \int \frac{1}{x} + x\,dx$

$\ln(y) + \frac{y^2}{2} = \ln(x) + \frac{x^2}{2} + C$

3. $\frac{dy}{dx} = \frac{x+y}{x}$

$y' = \frac{x+y}{x}$ $\qquad y = vx$

$y' = v \times 1 + x\frac{dv}{dx}$

$y' = \frac{x+vx}{x}$

$v + \frac{dv}{dx}x = \frac{x+vx}{x}$

$v + \frac{dv}{dx}x = 1 + v$

$\frac{dv}{dx} \times x = 1$

$\int dv = \int \frac{1}{x}\,dx$

$v = \ln(x) + C$

$v = \frac{y}{x}$

$\frac{y}{x} = \ln(x) + C$

$y = x\ln(x) + xC$

4. check for exactness & hence solve.

i) $(\cos x - y\sin x)dx + \cos x\,dy = 0$

$M = \cos x - y\sin x$ $\qquad N = \cos x$

$\frac{\partial M}{\partial y} = -\sin x$ $\qquad \frac{\partial N}{\partial x} = -\sin x$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$\therefore$ exact

$\int (\cos x - y\sin x)\,dx + \int 0 = C$

$+\sin x + y\cos x + 0 = C$

7) $(y+x^2)\,dx + x\,dy = 0 \qquad y(1) = 2$

$M = y + x^2 \qquad N = x$

$\frac{\partial M}{\partial y} = 1 + 0 \qquad \frac{\partial N}{\partial x} = 1$

$\therefore$ exact

$\frac{\partial N}{\partial y} = \frac{\partial N}{\partial x}$

$\int y + x^2\,dx + \int 0 = C$

$yx + \frac{x^3}{3} + 0 = C$

$yx + \frac{x^3}{3} = C$

$y = \frac{C}{x} - \frac{x^3}{3x}$

$y = \frac{C}{x} - \frac{x^2}{3}$

$y = C - \frac{x^2}{3}$

$y(1) = 2 \quad \therefore$

$C - \frac{1}{3} = 2$

$C = 2 + \frac{1}{3} = \frac{7}{3}$

8) $(2xy + y)\,dx + x^2\,dy = 0$

$M = 2xy + y \qquad N = x^2$

$\frac{\partial M}{\partial y} = 2x + 1 \qquad \frac{\partial N}{\partial x} = 2x$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \quad \therefore$ non-exact

$\int 2xy + y\,dx + \int 0 = 0$

$xy + \frac{x^2 y}{2} + 0 = C$

$xy = C$

$y = \frac{C}{x}$

$(2xy + y)\,dx + x^2\,dy = 0$

$y(2x+1)\,dx + x^2\,dy = 0$

$\frac{(2x+1)}{x^2}\,dx + \frac{1}{y}\,dy = 0$

$\int \frac{2x+1}{x^2}\,dx + \int \frac{1}{y}\,dy = 0$

$\int \frac{2}{x} + \frac{1}{x^2}\,dx + \int \frac{1}{y}\,dy = 0$

$2\ln(x) + \frac{-1}{x} + \ln y = \ln C$

$2\ln(x) - \frac{1}{x} + \ln y = \ln C$

$\ln\left(\frac{x^2 y}{C}\right) = \frac{1}{x}$

$y = \frac{C e^{1/x}}{x^2}$

$\frac{dy}{dx} + 2y = e^{-x}$

ii] $\frac{dy}{dx} + \frac{2}{x}y = x^2 \; ; \; y(1) = 1$ (Bernoulli)

$y' + 2y = e^{-x}$

$P = 2$

$\int P\,dx = 2x$

$e^{\int P\,dx} = e^{2x}$ $\Rightarrow$ IF

$e^{2x} \cdot y = \int e^{2x} \cdot e^{-x}\,dx$

$e^{2x} \cdot y = \int e^{x}\,dx$

$e^{2x} \cdot y = e^{x} + C$

$y = \frac{e^x}{e^{2x}} + \frac{C}{e^{2x}}$

$y = e^{x} \cdot e^{-2x} + C \cdot e^{-2x}$

$y = e^{-2x}(e^{x} + C)$

ii) $\frac{dy}{dx} + \frac{2}{x}y = x^2 \; ; \; y(1) = 1$

$y' + \frac{2}{x}y = x^2$

$P = \frac{2}{x}$

$\int P\,dx = \int \frac{2}{x}\,dx = 2\ln(x) = \ln(x^2)$

$e^{\int P\,dx} = e^{2\ln(x)} = e^{\ln(x^2)} = x^2$

$y \cdot x^2 = \int x^2 \cdot x^2\,dx$

$y \cdot x^2 = \int x^4 \cdot dx$

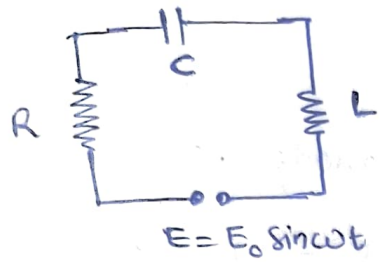
$x^2 y = \frac{x^5}{5} + C$

$y = \frac{x^5}{5x^2} + \frac{C}{x^2}$

$y = \frac{x^3}{5} + C$

5) Model ODE for R-L circuit and solve current $I(t)$ in emperess where 't' is the time

Modelling : RLC Circuit.



In R-L circuit,

$$LI' + RI = E(t) \quad \text{(by KVL)}$$

In RLC circuit, we are adding a voltage drop $\frac{Q}{C}$ across the capacitor where Q is the charge on the capacitor.

By Coulomb's law,

$$\boxed{I(t) = \frac{dQ}{dt}} \quad \text{or} \quad Q = \int I(t)\,dt$$

for RLC model, (by KVL), we have

$$LI' + RI + \frac{Q}{C} = E(t) \quad \text{—— (1)}$$

$$LI' + RI + \frac{1}{C}\int I\,dt = E(t)$$

differentiating

$$LI'' + RI' + \frac{1}{C}I = E'(t) \quad \text{—— (2)}$$

{(2) is the second order non-homogenous ODE with constant coefficients and solution of (2) gives the current at any time 't'}

(1) can be written as $I = Q'$

$$LQ'' + RQ' + \frac{1}{C}Q = E(t) \quad \text{—— (3)}$$

and the solution of (3) is Q(t).

eg:1 Find the steady state current in RLC model, where,

$R = 12\,\Omega$, $L = 0.4\,H$, $C = \frac{1}{50}\,F$, $E = 220\sin 10t\ V$

So, the model is;

$$LI'' + RI' + \frac{1}{C}I = E'(t)$$

$$0.4I'' + 12I' + 50I = 2200\sin 10t$$

$$I'' + 30I' + 200I = 5500\cos 10t \quad \text{——(a)}$$

To find $I_h(t)$ . . .

$$I'' + 30I' + 200I = 0$$

CE ; $\lambda^2 + 30\lambda + 200 = 0$

$$(\lambda+10)(\lambda+20) = 0$$

$\lambda_1 = -10$ $\lambda_2 = -20$ Case I

$$I_h(t) = C_1e^{-10t} + C_2e^{-20t}$$

To find $I_p(t)$

Take $I_p = A\cos 10t + B\sin 10t$

$I_p' = -10A\sin 10t + 10B\cos 10t$

$I_p'' = -100A\cos 10t - 100B\sin 10t$

(a) becomes

$-100A\cos 10t - 100B\sin 10t + 30(-10A\sin 10t + 10B\cos 10t) + 200(A\cos 10t + B\sin 10t) = 5500\cos 10t$

$\cos 10t\,(-100A + 300B + 200A) + \sin 10t\,(-100B - 300A + 200B) = 5500\cos 10t$,

equating coefficients .

$$100A + 300B = 5500 \quad \text{——(1)}$$

$$-300A + 100B = 0 \quad \text{——(2)}$$

(2)×3

$$100A + 300B = 5500$$

$$-900A + 300B = 0$$

$$1000A = 5500$$

$$\therefore A = 5.5$$

from (2) ;

$$B = 16.5$$

$$I_p(t) = 5.5\cos 10t + 16.5\sin 10t$$

∴ general solution ;

$$I(t) = C_1e^{-10t} + C_2e^{-20t} + 5.5\cos 10t + 16.5\sin 10t$$

The steady state soln of the model is obtained by letting $t \to \infty$

$e^{-\infty} \to 0$

As $t \to \infty$, $I(t) = 5.5 \cos 10t + 16.5 \sin 10t$

Q:- Find the steady state current in RLC model when R is negligibly very small.
$L = 0.5\,H$, $C = 0.005\,F$, $E = \sin t\ V$
Assume that the model has 0 initial current & charge

$R \approx 0$

$$LI'' + RI' + \frac{1}{C} I = E'(t)$$

$$0.5\, I'' + \frac{1}{0.005} I = \sin t + \cos t$$

$$I'' + 400 I = 2\cos t \quad \text{—①}$$

To find I_h ...

$$I'' + 400 I = 0$$

C.E $\quad m^2 + 400 = 0$

$$m^2 = -400$$

$$m = \pm 20i \quad \text{Case III.}$$

* $I_h = M\cos 20t + N \sin 20t$

To find I_p ...

choose $\quad I_p = A\cos t + B \sin t.$

$I_p' = -A\sin t + B\cos t$

$I_p'' = -A\cos t - B\sin t$

$I'' + 400 I = 2\cos t$

$(-A\cos t - B\sin t) + 400(A\cos t + B\sin t) = 2\cos t$

Equating Coefficients ...

$$-A + 400A = 2$$

$$399A = 2$$

$$A = \frac{2}{399}$$

$$-B + 400B = 0$$

$$B = 0$$

$$I_p = \frac{2}{399}\cos t$$

$\therefore$ $I(t) = M\cos 20t + N \sin 20t + \frac{2}{399}\cos t$ —②

$I(0) = 0$
$Q(0) = 0$ } given.

Apply $I(0) = 0$

$$0 = M + 0 + \frac{2}{399}$$

$$M = -\frac{2}{399} //$$

$I = Q'$

* $Q = \int I\,dt$

∴ we have;

Q(t) =

$$LI' + RI + \frac{1}{C}\underbrace{\int I\,dt}_{Q(t)} = E(t)$$

At $t = 0$;

$$LI'(0) + \frac{1}{C}Q(0) = E(0)$$

$$LI'(0) + 0 = 0$$

$$LI'(0) = 0$$

$$I'(0) = 0$$

Differentiating ② ;

$I'(t) =$

$$-20M\sin 20t + 20N\cos 20t - \frac{2}{399}\sin t$$

$$I'(0) = 0$$

$$0 = -20M \times 0 + 20N - 0$$

$$20N = 0$$

$$N = 0$$

∴ Current at t ;

$$I(t) = \frac{-2}{399}\cos 20t + \frac{20}{399}\cos t = \frac{2}{399}(\cos t - \cos 20t)$$

4- System Of ODE's

~~n~~ first order ODE.

A linear system of n first order ODE in n unknowns $y_1(t), y_2(t) \cdots y_n(t)$ is of the following form

$$\left.\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 + \cdots + a_{1n}y_n \\ y_2' &= a_{21}y_1 + a_{22}y_2 + \cdots + a_{2n}y_n \\ &\vdots \\ y_n' &= a_{n1}y_1 + a_{n2}y_2 + \cdots + a_{nn}y_n \end{aligned}\right\} \quad (1)$$

In particular for a two linear system of ODE, we have ;

$$\left.\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 \\ y_2' &= a_{21}y_1 + a_{22}y_2 \end{aligned}\right\} \quad (2)$$

eg: $y_1' = -5y_1 + 2y_2$

$y_2' = 13y_1 + \frac{1}{2}y_2$

The solution of ① are the function $y_1(t), y_2(t) \cdots y_n(t)$ that ~~sta~~ satisfies all the n eqns in ①.

The solutions can be considered as a vector $y(t)$

$$y(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{bmatrix}$$

from ②

$$\vec{y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix} e^{at} \\ \sin t \end{bmatrix}$$

$$y'(t) = \begin{bmatrix} y_1'(t) \\ y_2'(t) \end{bmatrix} \overset{eg}{=} \begin{bmatrix} ae^{at} \\ \cos t \end{bmatrix}$$

② becomes

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$y' = Ay \quad \longrightarrow ③$$

for ① ;

$$\vec{y}' = \begin{bmatrix} y_1' \\ y_2' \\ \vdots \\ y_n' \end{bmatrix} \quad A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}_{n \times n}$$

$$\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$\vec{y}'$ = In Eq ;

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} & 2 \\ 13 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Review on Matrices.

Triangular matrices.

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -0 & 5 & 0 & 0 \\ 2 & 1 & 4 & 0 \end{bmatrix}_{3\times 4}$$ lower triangular matrix.

$$B = \begin{bmatrix} & & \\ & & \\ & 0 & \end{bmatrix}$$

Elementary row operations:

1) $R_i \longleftrightarrow R_0$ (interchanging rows).

2) $R_i \longrightarrow cR_i$ (Scalar multiplication with a row)

3) $R_i \longrightarrow R_i + cR_j$ (row is added with a scalar multiplication of other row)

Gauss Elimination. {used to solve a system of linear eqns}

consider 3×3 system.

$$\left.\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}\right\}_{3\times 3}$$

$A\vec{x} = \vec{b}$

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

Augmented matrix :

$$(A/\vec{b}) = \left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ 0 & \star & \star & \star \\ 0 & + & + & + \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ 0 & \star & \star & \star \\ 0 & 0 & \star & \star \end{array}\right]$$

Solve by Gauss Elimination.

$x + y + 2z = 9$

$2x + 4y - 3z = 1$

$3x + 6y - 5z = 0$

$$A/\vec{b} = \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right]$$

$R_1 \Rightarrow$ pivot row.

$R_2 \Rightarrow 2R_1 - R_2 = (2\times1) - 2 = 0$ $\Rightarrow$ do all in row 2

$R_3 \Rightarrow 3R_1 - R_3 = (3\times1) - 3 = 0$ $\Rightarrow$ do all in row 3

$$= \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & -2 & 7 & 17 \\ 0 & -3 & 11 & 27 \end{array}\right]$$

$R_2 \Rightarrow$ pivot row

$R_3 \Rightarrow 3R_2 - 2R_3$

$3\times2 - -5 = 11$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & -2 & 7 & 17 \\ 0 & 0 & -1 & -3 \end{array}\right]$$

upper triangle form obtained.

$-1z = -3$

$z = 3$

$-2y + 7z = 17$

$-2y = -4$

$y = 2$

$x + 2 + 2\times3 = 9$

$x = 1$

$$\vec{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Eigen Value Problem:

Let A be $n \times n$ matrix

consider the equation;

$$A\vec{x} = \lambda\vec{x} \quad \text{—①}$$

$\vec{x}$ is $n \times 1$ unknown column vector of order $n \times 1$

A soln of ① is a non-zero vector $\vec{x}$ corresponding to the scalar λ.

λ is called the Eigen value of A and $\vec{x}$ is called the corresponding Eigen vector of A.

$$\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \lambda = 2$$

$$A\vec{x} = 2\vec{x}$$

for any $n \times n$ matrix, we can find 'n' eigen values say $\lambda_1, \lambda_2, \ldots, \lambda_n$ corresponding we can find

$\vec{x}_1, \vec{x}_2, \ldots \vec{x}_n$

$$A\vec{x} - \lambda\vec{x} = \vec{0}$$

$$(A - \lambda I)\vec{x} = \vec{0} \quad \text{—②}$$

Consider an eqn

$$|A - \lambda I| = 0 \quad \text{—③}$$

③ is called the characteristic eqn of A.

consider $(A - \lambda_i I)\vec{x}_i = \vec{0}$

Q: Find the eigen values & eigen vectors of the matrix

$$A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

Soln: The characteristic Eqn;

$$|A - \lambda I| = 0$$

$$A - \lambda I = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{vmatrix} = 0$$

$$\therefore (-5-\lambda)(-2-\lambda) - 4 = 0$$

$$10 + 2\lambda + 5\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 + 7\lambda + 6 = 0$$

$$(\lambda + 1)(\lambda + 6) = 0$$

$$\lambda_1 = -1 \qquad \lambda_2 = -6$$

⇒ Eigen Values.

To find $\vec{x}$ ....

take $\lambda_1 = -1$

$$(A - -1\,I)\,\vec{x_1} = \vec{0}$$

$$\begin{bmatrix} (-5--1) & 2 \\ 4 & (-2--1) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$R_2 \longrightarrow R_1 + 2R_2$

$$\begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-4x_1 + 2x_2 = 0$ (1 eqn 2 unknowns $\therefore$ 1 free choice)

Let $x_1 = t$

$-4t + 2x_2 = 0$

$2x_2 = 4t$

$x_2 = 2t$

$$\vec{x_1} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} t \\ 2t \end{bmatrix}$$

take $t = 1$

$$x_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$(A - -6I)\,\vec{x_2} = 0$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$R_2 \longrightarrow 2R_1 - R_2$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$x_1 + 2x_2 = 0$

$x_1 = -2x_1$

let $x_1 = t$

$x_2 = -2t$

take $t = 1$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} t \\ -2t \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

A linear system ODE is a 2×2 system of ODE ;

$$y_1' = a_{11}y_1 + a_{12}y_2$$

$$y_2' = a_{21}y_1 + a_{22}y_2$$

$$\vec{y}' = A\vec{y} \quad \text{—①}$$

eqn ① is called Homogeneous linear system of ODE.

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \quad \cdots$$

Homogeneous linear system of ODE with constant coefficients.

consider $\vec{y}' = A\vec{y}$, A has constant real entries

—①

Assume that ,

$\vec{y} = \vec{x}e^{\lambda t}$ is a soln of ①

$$\vec{y}' = \lambda\vec{x}e^{\lambda t}$$

① becomes

$$\lambda\vec{x}e^{\lambda t} = A\vec{x}e^{\lambda t}$$

$$A\vec{x} = \lambda\vec{x} \quad \text{(constitute an eigin value problem)}$$

—②

if A is an $n \times n$ matrix ② gives n eigin vectors. $\vec{x}^{(1)}, \vec{x}^{(2)}, \cdots \vec{x}^{(n)}$ corresponding to $\lambda_1, \lambda_2, \cdots \lambda_n$

then ,

∴ the independent solutions of ①

$$y_1 = \vec{x}^{(1)}e^{\lambda_1 t}$$

$$y_2 = \vec{x}^{(2)}e^{\lambda_2 t}$$

$$\vdots$$

$$y_n = \vec{x}^{(n)}e^{\lambda_n t}$$

$$\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$y = c_1\vec{x}^{(1)}e^{\lambda_1 t} + c_2\vec{x}^{(2)}e^{\lambda_2 t} + \cdots + c_n\vec{x}^{(n)}e^{\lambda_n t}$$

n-system ODE, general solution.

$$y = c_1\vec{x}^{(1)}e^{\lambda_1 t} + c_2\vec{x}^{(2)}e^{\lambda_2 t} + \cdots + c_n\vec{x}^{(n)}e^{\lambda_n t}$$

$$y_1(t) = \vec{x}^{(1)}e^{\lambda_1 t} \qquad y_2(t) = \vec{x}^{(2)}e^{\lambda_2 t}, \quad \cdots \quad y_n(t) = \vec{x}^{(n)}e^{\lambda_n t}$$

The wronskian of $y_1, y_2, \cdots, y_n$

$$W(y_1, y_2, \cdots, y_n) = \begin{vmatrix} x_1^{(1)}e^{\lambda_1 t} & \cdots & x_1^{(n)}e^{\lambda_n t} \\ x_2^{(1)}e^{\lambda_1 t} & \cdots & x_2^{(n)}e^{\lambda_n t} \\ \vdots & & \vdots \\ x_n^{(1)}e^{\lambda_1 t} & \cdots & x_n^{(n)}e^{\lambda_n t} \end{vmatrix}$$

where ;

$$\vec{x}^{(1)} = \begin{bmatrix} x_1^{(1)} \\ x_2^{(1)} \\ \vdots \\ x_n^{(1)} \end{bmatrix}$$

$$\vec{x}^{(n)} = \begin{bmatrix} x_1^{(n)} \\ x_2^{(n)} \\ \vdots \\ x_n^{(n)} \end{bmatrix}$$

$$\therefore \quad w = e^{\lambda_1 t + \lambda_2 t + \lambda_n t} \begin{vmatrix} x_1^{(1)} & \cdots & x_1^{(n)} \\ x_2^{(1)} & & x_2^{(n)} \\ \vdots & & \vdots \\ x_n^{(1)} & \cdots & x_n^{(n)} \end{vmatrix}$$

- if $w \neq 0$ for all 't', $y_i(t)$ are linearly independent $i = 1, 2, \ldots n$
- if $w = 0$ for at least one 't', y_i's are L.D.

∴ For a two system ODE, the solution,

$$\vec{y} = \begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix}$$

This represent two curves.

For each component of $y(t)$, this is a parametric rep with parameter 't'. This curves are called trajectories (orbit or path) the phase plane.

Eg: $y_1' = -3y_1 + y_2$

$y_2' = y_1 - 3y_2$

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \quad \vec{y}' = \begin{bmatrix} y_1' \\ y_2' \end{bmatrix},$$

$$A = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix}, \quad \vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\vec{y}' = A\vec{y} \quad \text{—①}$$

consider $A\vec{x} = \lambda\vec{x}$ (solve ①)

To find λ ;

The characteristic Eqn ;

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -3-\lambda & 1 \\ 1 & -3-\lambda \end{vmatrix} = 0$$

$$(-3-\lambda)^2 - 1 = 0$$

$$9 + \lambda^2 + 6\lambda - 1 = 0$$

$$\lambda^2 + 6\lambda + 8 = 0$$

$$(\lambda+2)(\lambda+4) = 0$$

$\lambda_1 = -2$ $\lambda_2 = -4$

To find $\vec{x}$ $(A-\lambda I)\vec{x} = 0$

Take $\lambda_1 = -2$

$(A - -2I)\vec{x}^{(1)} = 0$

$$\left(\begin{bmatrix}-3 & 1\\ 1 & -3\end{bmatrix} - \begin{bmatrix}-2 & 0\\ 0 & -2\end{bmatrix}\right)\begin{bmatrix}x_1\\ x_2\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\begin{bmatrix}-1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

$R_2 \longrightarrow R_1 + R_2$

$$\begin{bmatrix}-1 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

$-x_1 + x_2 = 0$

$x_1 = x_2$

$$\vec{x}^{(1)} = \begin{bmatrix}x_1\\ x_2\end{bmatrix} = \begin{bmatrix}x_1\\ x_1\end{bmatrix} = \begin{bmatrix}1\\ 1\end{bmatrix}$$

Take $\lambda_2 = -4$

$(A - -4I)\vec{x}^{(2)} = 0$

$$\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

$R_2 \longrightarrow R_1 - R_2$

$$\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

$x_1 + x_2 = 0$

$x_1 = -x_2$

$$\vec{x}^{(2)} = \begin{bmatrix}x_1\\ x_2\end{bmatrix} = \begin{bmatrix}x_1\\ -x_1\end{bmatrix} = \begin{bmatrix}1\\ -1\end{bmatrix}$$

$y_1(t) = x^{(1)}e^{-2t}$, $y_2(t) = \vec{x}^{(2)} e^{-4t}$

$\therefore$ general solution.

$$\vec{y}(t) = c_1\vec{x}^{(1)} e^{-2t} + c_2\vec{x}^{(2)} e^{-4t}$$

$$\begin{bmatrix}y_1(t)\\ y_2(t)\end{bmatrix} = \vec{y}(t) = c_1\begin{bmatrix}1\\ 1\end{bmatrix}e^{-2t} + c_2\begin{bmatrix}1\\ -1\end{bmatrix}e^{-4t}$$

$$\left.\begin{aligned} y_1(t) &= c_1 e^{-2t} + c_2 e^{-4t}\\ y_2(t) &= c_1 e^{-2t} - c_2 e^{-4t}\end{aligned}\right\} \text{①}$$

eqn ① represent the trajectories in $y_1 - y_2$ plane

In particular, if c_1 or c_2 is zero.

The trajectories represent two straight lines.

if c_1 or c_2 is zero.

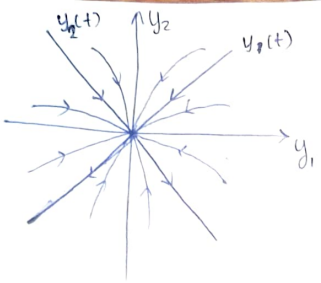
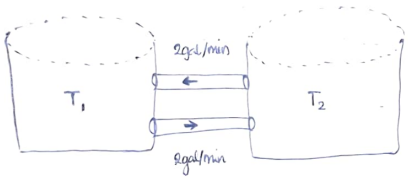


Fig: 2 Mixing Problem.

Tanks T_1 & T_2 contains initially 100 gal of water each. In T_1 the water is pure, whereas 150 lbs of fertilizer are dissolved in T_2. By circulating the liquid at a range of 2 gal/min and stirring the amount of fertilizer $y_1(t)$ in T_1 & $y_2(t)$ in T_2 change with time t. How long should we let the liquid circulate so that T_1 will contain at least half as much fertilizer as there'll be left in T_2.



System of Tanks.

Sol: At any time 't' the rate of change = Inflow/min - Outflow/min

In T_1, $y_1'(t) = \frac{2}{100} y_2 - \frac{2}{100} y_1$

In T_2,

$$y_2'(t) = \frac{2}{100} y_1 - \frac{2}{100} y_2$$

i.e.;

$$\left.\begin{aligned} y_1' &= -0.02 y_1 + 0.02 y_2 \\ y_2' &= 0.02 y_1 - 0.02 y_2 \end{aligned}\right\} \text{ 2×2 System of ODE.}$$

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} -0.02 & +0.02 \\ 0.02 & -0.02 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \Rightarrow \vec{y}' = A\vec{y}.$$

Solve $A\vec{x} = \lambda\vec{x}$

To find λ ;

$|A - \lambda I| = 0$

$$\begin{vmatrix} -0.02 - \lambda & 0.02 \\ 0.02 & -0.02 - \lambda \end{vmatrix} = 0$$

$$(-0.02 - \lambda)^2 - (0.02)^2 = 0$$

$$(-0.02)^2 + 0.04\lambda + \lambda^2 - (0.02)^2 = 0$$

$$\lambda^2 + 0.04\lambda = 0$$

$$\lambda(\lambda + 0.04) = 0$$

$$\lambda_1 = 0 \qquad \lambda_2 = -0.04$$

to find $\vec{x}_2$

Take $\lambda_1 = 0$ $\quad (A - 0 \times I) = 0$

$$\begin{bmatrix} -0.02 & 0.02 \\ 0.02 & -0.02 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$R_2 \longrightarrow R_1 + R_2$$

$$\begin{bmatrix} -0.02 & 0.02 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-0.02 x_1 + 0.02 x_2 = 0$$

$$-x_1 + x_2 = 0$$

$$\boxed{x_2 = x_1}$$

$$\vec{x}_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Take $\lambda_2 = -0.04$;

$$(A - (-0.04)I)\,\vec{x} = 0$$

$$\begin{bmatrix} 0.02 & 0.02 \\ 0.02 & 0.02 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$0.02x_1 + 0.02x_2$ $\quad R_2 \rightarrow R_1 - R_2$

$$\begin{bmatrix} 0.02 & 0.02 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$0.02x_1 + 0.02x_2 = 0$$

$$x_1 = -x_2$$

$$\vec{x}_2 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

∴ general solution

$$\vec{y}(t) = c_1 \vec{x}_1 e^{\lambda_1 t} + c_2 \vec{x}_2 e^{\lambda_2 t}$$

$$= c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{0t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-0.04t}$$

$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix} c_1 + c_2 e^{-0.04t} \\ c_1 - c_2 e^{-0.04t} \end{bmatrix}$$

* Initial condition ;

$$y_1(0) = 0 \;, \quad y_2(0) = 150$$

$$y_1(t) = c_1 + c_2 e^{-0.04t}$$

$$0 = c_1 + c_2 \quad -(1)$$

$$y_2(0) = 150$$

$$y_2(t) = c_1 - c_2 e^{-0.04t}$$

$$150 = c_1 - c_2 \quad -(2)$$

$$c_1 + c_2 = 0 \quad +$$

$$c_1 - c_2 = 150$$

$$2c_1 = 150$$

$$c_1 = 75$$

$$c_2 = -75$$

$y_1(t) = 75 - 75e^{-0.04t}$

$y_2(t) = 75 + 75e^{-0.04t}$

To find 't' such that,

$y_1(t) = \frac{1}{2} y_2(t)$

Take $y_2 = 150 - y_1$ (amnt left in T_2)

$y_1 = \frac{1}{2}(150 - y_1)$

$y_1 = 75 - \frac{y_1}{2}$

$2y_1 = 150 - y_1$

$3y_1 = 150$

$y_1 = 50$

To find 't', we have

$y_1 = 50$

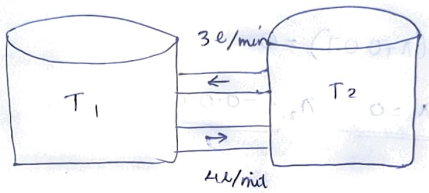
$75 - 75e^{-0.04t} = 50$

$e^{-0.04t} = 1/3$

$-0.04t = -\ln 3$

$t = \frac{\ln 3}{0.04} = 27.5 \text{ min}$

Q: 2 Tanks T_1 & T_2 initially containing 100 l of water. Tank T_1 initially contains 10kg of salt, & Tank T_2 contains 20kg of salt. The liquid is pumped from T_1 to T_2 at the rate of 4l/min & the same time liq flows from T_2 to T_1 at the rate of 3 litre per min. The mixtures in both tanks are well stirred. Find the formulate the system of ODE & determine the amnt of salt in each tank at any time 't'.



model:

$y_1'(t) = \frac{3}{100} y_2 - \frac{4}{100} y_1 = -0.04 y_1 + 0.03 y_2$

$y_2'(t) = \frac{4}{100} y_1 - \frac{3}{100} y_2 = 0.04 y_1 - 0.03 y_2$

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} -0.04 & 0.03 \\ 0.04 & -0.03 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Solve $A\vec{x} = \lambda\vec{x}$

To find λ ...

$|A - \lambda I| = 0$

$$\begin{vmatrix} -0.04-\lambda & 0.03 \\ 0.04 & -0.03-\lambda \end{vmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left[(-0.04-\lambda)(-0.03-\lambda)\right] - \left[(0.03)(0.04)\right] = 0$$

$$(-0.04\lambda - 0.03) + (0.07\lambda) + \lambda^2 - (0.04 \cdot 0.03) = 0$$

$$\lambda^2 + 0.07\lambda = 0$$

$$\lambda(\lambda + 0.07) = 0$$

$$\lambda_1 = 0 \qquad \lambda_2 = -0.07$$

To find $\vec{x}$; $\lambda_1 = 0$

$$A\vec{x} = \lambda\vec{x}$$

$$\begin{bmatrix} -0.04 & 0.03 \\ 0.04 & -0.03 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$R_2 \rightarrow R_1 + R_2$$

$$\begin{bmatrix} -0.04 & 0.03 \\ 0 & 0.00 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$-0.04x_1 + 0.03x_2 = 0$$

$$0.03x_2 = 0.04x_1$$

$$x_2 = \frac{4}{3}x_1$$

$$\therefore \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ \frac{4}{3}x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{4}{3} \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

For $\lambda_2 = -0.07$.

$$\begin{bmatrix} -0.04 & 0.03 \\ 0.04 & -0.03 \end{bmatrix}$$

Reduction of an n^{th} order ODE to a system

$y' = f(t, y)$

$y'' = f(t, y', y)$

$y''' = f(t, y'', y', y)$

The n^{th} order ODE can be written in the following ~~order~~ form.

$$y^{(n)} = f(t, y, y', y'', \cdots, y^{(n-1)}) \quad \text{——} (1)$$

eqn (1) can be converted to a system of n - first order ODE by setting $y_1 = y$, $y_2 = y'$, $y_3 = y''$, $y_n = y^{(n-1)}$

Then the system of ODE is given by,

$$y_1' = y_2$$

$$y_2' = y_3$$

$$y_{n-1}' = y_n$$

$$y_n' = F(t, y_1, y_2, \cdots, y_n)$$

eg:- Convert the following ODE into a system of first order linear ODE and hence, solve;

$$4y'' - 15y' - 4y = 0$$

$y_1 = y$, $y_2 = y'$, $y_3 = y''$

$y_1' = y_2 = 0y_1 + y_2 + 0y_3$

$y_2' = y_3 = y'' = \frac{15}{4} y' + y$

$4y'' = 15y' + 4y$

$y'' = \frac{15}{4} y' + y$

$y_2' = y_1 + \frac{15}{4} y_2 + y_3$ → ~~$0y_1 + y_2 + 0y_3 + y_1 + \frac{15}{4} y_2$~~

$y_1' = 0y_1 + y_2 + 0y_3$

$y_2' = y_1 + 15y_2 + 0y_3$

$$\left.\begin{aligned} y_1' &= 0y_1 + y_2 \\ y_2' &= y_1 + \tfrac{15}{4} y_2 \end{aligned}\right\} \text{ 2x2 system of ODE.}$$

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 15/4 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Solve $A\vec{x} = \lambda\vec{x}$

To find λ

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 0-\lambda & 1 \\ 1 & \frac{15}{4} - \lambda \end{vmatrix} = 0$$

$$-\lambda\left(\frac{15}{4} - \lambda\right) - 1 = 0$$

$$-\frac{15}{4}\lambda + \lambda^2 - 1 = 0$$

$$4\lambda^2 - 15\lambda - 4 = 0$$

$$4\lambda^2 - 16\lambda + \lambda - 4 = 0$$

$$4\lambda(\lambda - 4) + (\lambda - 4) = 0$$

$$(\lambda - 4) + (4\lambda + 1) = 0$$

$\lambda_1 = 4$ $\quad \lambda_2 = -1/4$

To find $\vec{x}$.

$\lambda_1 = 4$

$(A - 4I)\vec{x} = 0$

$$\begin{bmatrix} -4 & 1 \\ 1 & -1/4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$R_2 \to R_1 + 4R_2$

$$\begin{bmatrix} -4 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-4x_1 + x_2 = 0$

$x_2 = 4x_1$

$$\vec{x}_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

To find $\vec{x}_2$

$\lambda_2 = -1/4$

$\left(A + \frac{I}{4}\right)\vec{x} = 0$

$$\begin{bmatrix} 1/4 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$R_2 \to 4R_1 - R_2$

$$\begin{bmatrix} 1/4 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\frac{1}{4}x_1 + x_2 = 0$

$x_1 = -4x_2$

$$\vec{x}_2 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -4x_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$$

The general solution solution;

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = c_1\vec{x}_1 e^{\lambda_1 t} + c_2\vec{x}_2 e^{\lambda_2 t}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} -4 \\ 1 \end{bmatrix} e^{-\frac{1}{4}t}$$

$y_1 = c_1 e^{4t} + 4c_2 e^{-4t}$

$y_2 = c_2 e^{-4t} - 4c_1 e^{4t}$

∴ Solution to the given 2nd ODE.

$y = y_1 = c_1 e^{4t} - 4c_2 e^{-\frac{1}{4}t}$

2. $y''' + 2y'' - y' - 2y = 0$

$y_1 = y \;,\; y_2 = y' \;,\; y_3 = y'' \;,\; y_4 = y'''$

$y_1' = y_2$

$y_2' = y_3$

$y_3' = y''' = -2y'' + y' + 2y = -2y_3 + y_2 + 2y_1$

$$\begin{bmatrix} y_1' \\ y_2' \\ y_3' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \Rightarrow \vec{y}' = A\vec{y}$$

3×3 system of ODE.

Solve $A\vec{x} = \lambda\vec{x}$

$|A - \lambda I| = 0$

$$\begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 2 & 1 & -2-\lambda \end{vmatrix} = 0$$

$-\lambda(-\lambda(-2-\lambda) - 1) - 1(0 - 2) + 0 = 0$

$-\lambda(2\lambda + \lambda^2 - 1) + 2 = 0$

$-2\lambda^2 + \lambda^3 + 2 + \lambda = 0$

$-2\lambda^2 - \lambda^3 + \lambda + 2 = 0$

$2\lambda^2 + \lambda^3 - \lambda - 2 = 0$

$\lambda^3 + 2\lambda^2 - \lambda - 2 = 0$

$\lambda - 1$ is root, $(\lambda - 1)$ is a factor

$$(\lambda - 1) \overline{)\ \lambda^3 + 2\lambda^2 - \lambda - 2}$$

$\lambda^2 + 3\lambda + 2$

$\lambda^3 - \lambda^2$ (−)

$3\lambda^2 - \lambda$

$3\lambda^2 - 3\lambda$ (−)

$2\lambda - 2$

$2\lambda - 2$ (−)

0

$\lambda^2 + 3\lambda + 2$ is other factor.

$(\lambda + 1)(\lambda + 2)$

$(\lambda - 1)(\lambda + 1)(\lambda + 2) = 0$

$\lambda_1 = 1 \;,\; \lambda_2 = -1 \;,\; \lambda_3 = -2$

eigen value of A.

3- Higher Order Linear ODE.

n^{th} order linear ODE.

$$y^n + P_{n-1}(x)y^{(n-1)} + \cdots + P_1(x)y' + P_0(x)y = r(x) \quad \text{—①}$$

$P_0, P_1, \ldots, P_{(n-1)}$ and r are f^n of x and are constant.

If $r(x) = 0$, then ① becomes,

$$y^n + P_{n-1}(x)y^{n-1} + \cdots + P_1(x)y' + P_0(x)y = 0$$

A Solution of ① is a f^n $y = h(x)$ that is defined and it is n times differentiate, when we replace that eqn ① satisfies, when we replace $y, y', \ldots y^n$ by $h, h', \ldots h^n$

A general Sol^n of the given eqn is in the following form:

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + \cdots + C_n y_n(x)$$

where $y_1(x), y_2(x), \ldots, y_n(x)$ are the basis of Sol^n and $C_1, C_2, \ldots, C_n$ are arbitrary constant.

1. If $P_0, P_1, \ldots, P_{(n-1)}$ and r are constant then eqn ①, ② called the n^{th} order linear ODE with constant coefficient.

2. IVP: an IVP for eqn ② contains 'n' initial conditions given by:

$$y(x_0) = k_0, \; y'(x_0) = k_1, \ldots, \; y^{(n-1)}(x_0) = k_{(n-1)}$$

Eg: ① Solve $y^{IV} - 5y'' + 4y = 0$

g] Solve IVP:

$$y''' - y'' + 100y' - 100y = 0 \quad , y(0) = 4 \quad , y'(0) = 11$$

$$y''(0) = -299.$$

Soln: The CE;

$$\lambda^3 - \lambda^2 + 100\lambda - 100 = 0$$

By inspection; $\lambda = 1$ is a root

$$\lambda - 1 \;\Big|\; \lambda^3 - \lambda^2 + 100\lambda - 100$$

$$\lambda^3 - \lambda^2$$

$$0 \quad 100\lambda - 100$$

$$100\lambda - 100$$

$$0$$

$$\lambda^2 + 100 = 0$$

$$\lambda^2 = -100$$

$$\lambda = \pm 10i$$

The roots are

$$\lambda_1 = 1 \quad , \lambda_2 = 10i \quad , \lambda_3 = -10i$$

Case ① λ $e^{\lambda x}$

Case III $\alpha + i\beta$ $e^{\alpha x}\cos\beta x$

$\therefore$ The general solution;

$$y = c_1 e^x + A\cos 10x + B\sin 10x$$

$$y(0) = 4.$$

$$c_1 + A + 0 = 4$$

$$A + c_1 = 4 \quad -①$$

$$y' = c_1 e^x - 10A\sin 10x + 10B\cos 10x$$

$$y'(0) = 11$$

$$11 = c_1 + 10B \quad -②$$

$$y'' = c_1 e^x - 100A\cos 10x - 100B\sin 10x$$

$$y''(0) = -299$$

$$-299 = c_1 - 100A \quad -③$$

① $C_1 - 100A = -299$

$C_1 + A = 4$ (−)

$-101A = -303$

$A = \frac{303}{101} = 3$

② $C_1 + 10B = 11$

$10B = 11 - 1$

$B = \frac{10}{10} = 1$

③ $C_1 + 3 = 4$

$C_1 = 1$

$\therefore y = e^x + 3\cos 10x + \sin 10x$

⇒ Euler - Cauchy in Higher Order.

Q: Solve

$x^3y''' - 3x^2y'' + 6xy' - 6y = 0$, $y(1) = 2$, $y'(1) = 1$, $y''(1) = 4$

let $y = x^m$ be a solution.

$y' = mx^{m-1}$

$y'' = m(m-1)x^{(m-2)}$

$y''' = m(m-1)(m-2)x^{(m-3)}$

Eqn becomes;

$x^3 \cdot m(m-1)(m-2)x^{(m-3)} - 3x^2 m(m-1)x^{(m-2)} + 6x\,mx^{m-1} - 6x^m = 0$

$m(m-1)(m-2)x^m - 3m(m-1)x^m + 6mx^m - 6x^m = 0$

$\left(m(m-1)(m-2) - 3m(m-1) + 6m - 6\right)x^m = 0$, $x^m \neq 0$

$m(m-1)(m-2) + 3m(m-1) - 6m - 6 = 0$

$(m-1)\{\, m(m-2) - 3m + 6 = 0$

$(m-1)(m^2 - 2m - 3m + 6 = 0$

$(m-1)\{m^2 - 5m + 6 = 0$

$m - 1 = 0$

$m_1 = 1$

$m^2 - 5m + 6 = 0$

$(m-2)(m-3) = 0$

$\therefore m_2 = 2$ $m_3 = 3$

$m = 1, 2, 3$ Case I ;

Basis of soln ; x^1, x^2, x^3.

the general soln is ;

$y = C_1x + C_2x^2 + C_3x^3$

$y(1) = ?$

$2 = c_1 + c_2 + c_3 \quad -①$

$y' = c_1 x + 2c_2 x + 3c_3 x^2$

$y'(1) = 1$

$1 = c_1 + 2c_2 + 3c_3 \quad -②$

$y'' = c_1 + 2c_2 + 6c_3 x$

$y''(1) = -4$

$-4 = c_1 + 2c_2 + 6c_3 \quad -③$

$c_1 + 2c_2 + 6c_3 = -4$
$c_1 + 2c_2 + 3c_3 = 1$
$-$

$3c_3 = -5$

$c_3 = -\frac{5}{3}$

② - ①

$c_1 + 2c_2 + 3c_3 = 1$
$c_1 + c_2 + c_3 = 2$
$-$

$c_2 + 2c_3 = -1 \quad -③$

$2c_2 + 6c_3 = 3$
$2c_2 + 6$

$c_1 + c_2 + c_3 = 2$
$c_1 + 2c_2 + 3c_3 = 1$
$\quad 2c_2 + 6c_3 = -4$

} 3×3 system.

$$(A/b) = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 1 \\ 0 & 2 & 6 & 4 \end{bmatrix}$$

$R_2 \to R_2 - R_1$

$$= \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 2 & 6 & 4 \end{bmatrix}$$

$R_3 \to 2R_2 - R_3$

$$= \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & -2 & 2 \end{bmatrix}$$
$c_1 \quad c_2 \quad c_3$

$-2c_3 = 2$

$c_3 = -1$

$c_2 + 2c_3 = -1$

$c_2 = -1 + 2$

$c_2 = 1$

$c_1 + c_2 + c_3 = 2$

$c_1 + 1 - 1 = 2$

$c_1 = 2$

$$\therefore \quad y = 2x + x^2 - x^3$$

→ Higher order non-Homogeneous ~~linear~~ ODE. with Constant Coeff

The general form is n^{th} order;

$$y^{(n)} + P_{(n-1)} y^{(n-1)} + \cdots + P_1 y' + P_0 y = r(x) \quad \text{—— (1)}$$

$r(x) \neq 0$, P_{n-1}, P_{n-2}, ..., P_1, P_0 are constants

The general soln of (1) is given by;

$$y = y_h + y_p$$

where, y_h is the general solution of the corresponding homogeneous ODE of (1).

y_p is any particular soln of (1).